EXAM PACK

EEE329

Basic Communication Engineering



Exam Pack

Session 2019–20 *Solutions and formula sheet of the 3rd Year 2nd Semester Final Examination-2022*

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Prepared by Nur Mohammad

November 2026

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PAST PAPER SOLUTIONS

EEE329

Basic Communication Engineering

Question Paper Solutions

Session 2019–20

EEE329 3rd Year 2nd Semester Final Examination-2022

QUESTION MAP

Q1	Q2	Q3	Q4	Q5	Q6
25 marks	25 marks	25 marks	25 marks	25 marks	25 marks

EACH ANSWER

Theory → Answer → Keyword Explanation → Notes and Remarks → Final Concise Answer

Prepared by Nur Mohammad

November 2026

BACK OF THE COVER

The paper at a glance

3 h	100	4 of 6	45 min
Total time	Total marks	Questions to answer	Per question

Time = marks × 1.8 min (180 min for 100 marks). Tick the questions you will answer; each sub-part links to its solution.

Q	Sub-part	Marks	Time	Pick
Part-A · Answer any two of the following questions				
Q1	(a) Short Notes: Twisted-Pair Cable and Radio Waves	6	11 min	☐
	(b) DSB-SC Spectra for Three Messages	6	11 min	
	(c) Shannon Capacity and the SNR for Double the Rate	6	11 min	
	(d) Equivalent Noise Resistance of Amplifiers in Cascade	7	13 min	
Q2	(a) Average Information per Message Equals the Entropy	8	14 min	☐
	(b) Power, Deviation, Deviation Ratio and Bandwidth of an Angle-Modulated Signal	8	14 min	
	(c) Effect of Phase and Frequency Error in Coherent DSB-SC Detection	6	11 min	
	(d) Condition on the Message Spectrum for SSB by Filtering	3	5 min	
Q3	(a) Phase-Shift Method of SSB Generation	8	14 min	☐
	(b) Effect of the Message Amplitude and Frequency on the FM Spectrum and Bandwidth	8	14 min	
	(c) Design of an Armstrong FM Modulator for 97.3 MHz	9	16 min	
Part-B · Answer any two of the following questions				
Q4	(a) FM Modulation Index for Three Settings	5	9 min	☐
	(b) Slope Detection of FM	6	11 min	
	(c) Uniform and Non-Uniform Quantization	4	7 min	
	(d) Data Rate and Bandwidth for Five Telemetry Signals	5	9 min	
	(e) The DM Receiver is an Accumulator	5	9 min	
Q5	(a) Sigma-Delta Modulation and its Advantages over DM	5	9 min	☐
	(b) Line Coding: Desired Properties, and Manchester, AMI and Dicode for ‘1010011100’	7	13 min	
	(c) DPSK Principle and Block Diagram	8	14 min	
	(d) Coherent Detection of BFSK	5	9 min	
Q6	(a) QPSK Modulation and its Advantages over BPSK	8	14 min	☐
	(b) TDM Output, Frame Rate and Bit Rate for Four 100-kbps Channels	6	11 min	
	(c) Frequency-Division Multiplexing	5	9 min	
	(d) Short Notes: Pulse Stuffing and Multiple-Slot Multiplexing	6	11 min	

Formulas and conventions follow Lathi and Ding, Modern Digital and Analog Communication Systems (4th ed.); 1(a), 1(c), 6(b), 6(c) and 6(d) follow Forouzan, Data Communications and Networking; 1(d) follows Kennedy and Davis, Electronic Communication Systems; the dicode rule in 5(b) follows the course notes.

Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering

3rd Year 2nd Semester Final Examination-2022

Session: 2019-20

Course Code: EEE 329 **Course Title:** Basic Communication Engineering

Total Marks: 100 **Credit: 3.0** **Total Time: 3 Hours**

Part-A

[Answer any two of the following questions]

1. (a) Write short notes on the following communication channels:

   (i) Twisted pair cable
   (ii) Radio wave 6

   (b) For carrier frequency 100 kHz, find the frequency, sketch the spectrum, indicate the upper sideband, lower sideband and their amplitudes of the following signals.

   (i) $m(t) = 1 + \cos(377t)$
   (ii) $m(t) = 2\sin^2(100\pi t)$
   (iii) $m(t) = 3\cos(1000t)\cos(3000t)$ 6

   (c) States Shannon's capacity theorem. A user wishes to transmit data over a channel having bandwidth B = 3000 Hz. If the SNR is 9.08 find the theoretical maximum bit rate of the channel. Also find the required SNR if the desired bitrate of the channel is doubled. 6

   (d) Find the equivalent noise resistance for several amplifiers in cascade. 7

2. (a) Show that the average information per message is equal to the entropy of the information source. 8

   (b) An angle modulated signal with carrier frequency 10 MHz is described by the equation $s(t) = 12\cos(\omega_c t + \sin 3000\pi t + 10\sin 2000\pi t)$

   (i) Find the power of the modulated signal
   (ii) Find the frequency deviation Δf
   (iii) Find the β
   (iv) Find the bandwidth. 8

   (c) Explain the effect of phase and frequency error during coherent detection of DSB-SC. 6

   (d) Describe the condition on message signal spectrum for SSB generation using selective filtering. 3

3. (a) Using proper block diagram explain phase shifting method of SSB generation. 8

   (b) Explain the effect of variation of amplitude and frequency of message signal on the spectrum and bandwidth of FM. 8

   (c) Design an Armstrong indirect FM modulator to generate an FM signal with carrier frequency 97.3 MHz and $\Delta f = 10.24\ kHz$. A NBFM generator of $f_{c1} = 20\ kHz$ and $\Delta f = 5\ Hz$ is available. Only frequency doublers can be used as multipliers. Additionally, a local oscillator (LO) with adjustable frequency between 400 and 500 kHz is readily available for frequency mixing. 9

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Part-B

[Answer any two of the following questions]

4. (a) In an FM system the frequency deviation is 5 kHz, when its audio modulating frequency is 400 Hz, and the audio modulating voltage is 2 V. Compute the modulation index. Also compute the frequency deviation and the modulation index if (i) if voltage is increased to 6V keeping modulation frequency unaltered, (ii) if voltage is increased to 8 V while the modulating frequency is reduced to 200 Hz. 5

   (b) Explain slope detection method of FM demodulation. 6

   (c) Define and explain uniform and non-uniform quantization. 4

   (d) Five telemetry signals, each of bandwidth 2400 Hz, are to be transmitted simultaneously by binary PCM. The signals must be sampled at least 20% above the Nyquist rate. Framing and synchronizing requires an additional 0.5% extra bits. A PCM encoder is used to convert these signals before they are time-multiplexed into a single data stream. Determine the minimum possible data rate (bit per second) that must be transmitted, and the minimum bandwidth required to transmit the multiplexed signal. 5

   (e) Show that the receiver of DM is essentially an accumulator. 5

5. (a) Explain principle of sigma-delta modulator. Also explain how it is advantageous over DM. 5

   (b) Briefly explain the desired characteristics of a line coding scheme. Draw the line coded signal for the information bits '1010011100' for the following line coding scheme- i) Manchester, ii) AMI, iii) Di-code. 7

   (c) Explain DPSK principle with necessary block diagram. 8

   (d) Describe coherent detection of BFSK. 5

6. (a) With proper block diagram, describe modulation of QPSK. Also explain how it offers advantages over BPSK? 8

   (b) A multiplexer combines four 100-kbps channels using a time slot of 2 bits. Show the output for the following inputs- '110010', '001010', '101101', '000111'. What is the frame rate? What is the frame duration? What is the bit rate? What is the bit duration? 6

   (c) Briefly explain frequency division multiplexing techniques. 5

   (d) Write short notes on following terms-

   (i) Pulse stuffing
   (ii) Multi-slot multiplexing 6

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1. (a) Write short notes on the following communication channels:

   (i) Twisted pair cable
   (ii) Radio wave Marks: 6

   (b) For carrier frequency 100 kHz, find the frequency, sketch the spectrum, indicate the upper sideband, lower sideband and their amplitudes of the following signals.

   (i) $m(t) = 1 + \cos(377t)$
   (ii) $m(t) = 2\sin^2(100\pi t)$
   (iii) $m(t) = 3\cos(1000t)\cos(3000t)$ Marks: 6

   (c) States Shannon's capacity theorem. A user wishes to transmit data over a channel having bandwidth B = 3000 Hz. If the SNR is 9.08 find the theoretical maximum bit rate of the channel. Also find the required SNR if the desired bitrate of the channel is doubled. Marks: 6

   (d) Find the equivalent noise resistance for several amplifiers in cascade. Marks: 7

1.(a) Write short notes on the following communication channels: (i) Twisted pair cable (ii) Radio wave Marks: 6

1. Theory

1. What is a communication channel? (Core Idea)

A **channel** is the path a signal takes from the transmitter to the receiver. Forouzan calls it the **transmission medium** and splits it into two families:

- **Guided media** (wired): the signal is led along a physical path. Examples: twisted-pair cable, coaxial cable, optical fibre.
- **Unguided media** (wireless): the signal travels as an electromagnetic wave through air or space, with nothing to hold it. Examples: radio waves, microwaves, infrared.

Every channel has three properties a designer cares about: how much **bandwidth** it offers (how fast it can carry data), how much it **attenuates** (weakens) the signal per kilometre, and how much **noise and interference** it lets in.

2. Why twist two wires?

A pair of wires carries a signal as the **difference** between the voltages on the two wires. Nearby equipment and other cables put unwanted noise voltage onto the wires. If one wire sat closer to the noise source, it would pick up more noise than the other, and the difference (the signal the receiver reads) would be corrupted. Twisting the pair swaps the positions of the two wires every half twist, so over a short length each wire is as often near the noise as the other. Both wires then pick up nearly the same noise, and it cancels when the receiver takes the difference.

3. What is a radio wave?

A radio wave is an electromagnetic wave: an electric and a magnetic field that travel together at the speed of light, $c = 3 \times 10^8$ m/s. Its wavelength is $\lambda = c/f$. Forouzan calls the waves from **3 kHz to 1 GHz** radio waves (above that, up to 300 GHz, are microwaves). The long wavelengths of radio waves give them two useful habits: they spread out in all directions from a simple antenna, and they pass through walls.

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2. Answer

(i) Twisted-pair cable

A twisted-pair cable is two insulated copper conductors twisted around each other. One wire carries the signal, the other is the ground reference, and the receiver uses the **difference** between them.

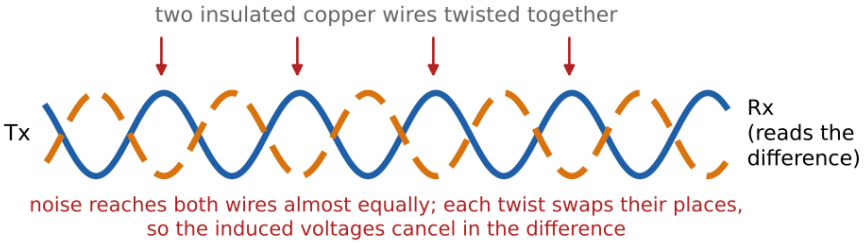


Figure 1: Twisted pair: each twist swaps the wires, so interference (red arrows) reaches both equally and cancels in the difference

Reading the figure. The solid blue and the dashed orange wires wind around each other from the transmitter (Tx) to the receiver (Rx). The red arrows are interference from outside. Because the wires keep changing places, each one gets the same share of the interference, and the receiver, which reads only the difference, sees it cancel.

- **Why twisted:** noise and crosstalk are induced almost equally in both wires and cancel at the receiver; the more twists per metre, the better the cancellation.
- **Types:** **UTP** (unshielded twisted pair), the common cheap kind, and **STP** (shielded twisted pair), which adds a metal foil or braid around the pair to block interference, at higher cost and weight.
- **Categories:** the EIA grades UTP into categories (Cat 1 to Cat 7). Higher categories have more twists and better insulation, so they carry higher data rates (Cat 5e about 100 Mbps to 1 Gbps; Cat 6 up to 10 Gbps over short runs).
- **Performance:** attenuation rises sharply with frequency, so long runs need repeaters, and the usable bandwidth is limited (a few hundred MHz at most).
- **Connector:** RJ45.
- **Uses:** telephone lines (the local loop), DSL, and Ethernet local-area networks (10Base-T, 100Base-T, 1000Base-T).
- **Pros and cons:** cheap, light and easy to install; but limited bandwidth and distance, and more sensitive to noise than coaxial cable or fibre.

(ii) Radio waves

Radio waves are electromagnetic waves between about **3 kHz and 1 GHz** used as an unguided channel.

- **Propagation:** they are mostly **omnidirectional**: an antenna sends them in all directions, so the sending and receiving antennas need not be aligned. Depending on frequency they travel as **ground waves** (below about 2 MHz, following the Earth's curve), **sky waves** (about 2 to 30 MHz, bent back to Earth by the ionosphere) or **line-of-sight** waves (above about 30 MHz).

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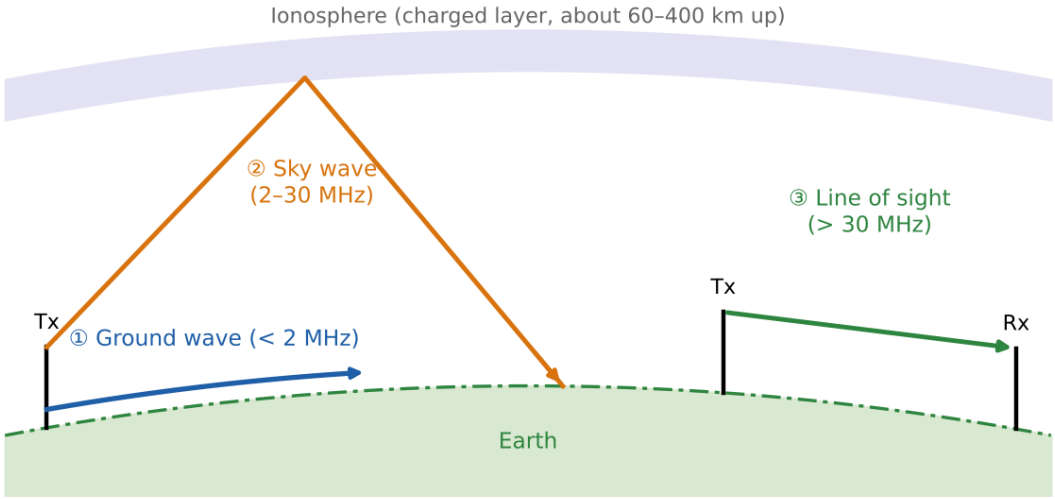


Figure 2: The three ways a radio wave travels (reused from the 2021-22 paper)

Reading the figure. Wave 1 (blue) hugs the curved Earth (ground wave). Wave 2 (orange) goes up, is bent back by the ionosphere and lands far away (sky wave). Wave 3 (green) goes straight from the Tx tower to the Rx tower (line of sight).

- **Penetration:** low and medium frequencies pass through walls, so a receiver inside a building still gets the signal. This is an advantage (no need to be outdoors) and a disadvantage (signals cannot be kept inside one building).
- **Interference and sharing:** because everyone shares the same air, interference between users is possible, and the frequency bands are licensed and controlled by government authorities (except some unlicensed bands).
- **Bandwidth:** the radio band is narrow (under 1 GHz in total), so data rates are low compared with microwaves.
- **Antennas:** omnidirectional antennas; the size of an antenna is comparable to the wavelength, so low frequencies need large antennas.
- **Uses:** one-to-many (**multicast**) communication: AM and FM radio broadcasting, television, maritime radio, cordless phones and paging.

Final Summary: twisted pair = cheap guided medium, noise cancelled by twisting, used for telephone and LAN, limited bandwidth and distance; radio waves = unguided, omnidirectional, 3 kHz to 1 GHz, travel by ground, sky or line of sight and pass through walls, used for broadcasting.

3. Keyword Explanation

1. Crosstalk

↩ back

Unwanted coupling of a signal from one pair of wires into a neighbouring pair. Twisting reduces it.

2. Omnidirectional antenna

↩ back

An antenna that radiates (or receives) roughly equally in all horizontal directions.

4. Notes and Remarks

- Forouzan (Ch. 7) treats twisted pair, coaxial cable and fibre as guided media, and radio waves, microwaves and infrared as unguided media; the frequency limits above are his.
- Microwaves (1 to 300 GHz) are **unidirectional**: they need dish or horn antennas pointed at each other, and they do not pass through walls. That is the main contrast with radio waves.

1.(b) For carrier frequency 100 kHz, find the frequency, sketch the spectrum, indicate the upper sideband, lower sideband and their amplitudes of the following signals. (i) $m(t) = 1 + \cos(377t)$ **(ii)** $m(t) = 2\sin^2(100\pi t)$ **(iii)** $m(t) = 3\cos(1000t)\cos(3000t)$ **Marks: 6**

1. Theory

1. What does multiplying by a carrier do? (Core Idea)

In **DSB-SC** (double-sideband suppressed-carrier) modulation the message $m(t)$ is simply multiplied by the carrier:

$$\varphi(t) = m(t)\cos\omega_c t, \qquad \omega_c = 2\pi f_c$$

Take one message tone, $m(t) = a\cos\omega_m t$. Use the identity $\cos A\cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$ with $A = \omega_c t$ and $B = \omega_m t$:

$$a\cos\omega_m t\cos\omega_c t = \frac{a}{2}\cos(\omega_c - \omega_m)t + \frac{a}{2}\cos(\omega_c + \omega_m)t$$

So one tone at f_m becomes *two* tones, at $f_c - f_m$ and $f_c + f_m$, each with *half* the amplitude. The tone above the carrier belongs to the **upper sideband (USB)**, the one below to the **lower sideband (LSB)**.

2. What about a constant (dc) part of the message?

If $m(t)$ contains a constant c, then $c\cos\omega_c t$ is a line **at the carrier frequency itself**, with amplitude c (not halved). A dc part of the message therefore shows up as a carrier.

3. How to read a spectrum

A (one-sided) amplitude spectrum draws one vertical line for each cosine in the signal, at its frequency, with the line's height equal to its amplitude. In the two-sided spectrum of Lathi, every cosine of amplitude a splits into two lines of height $a/2$ at $\pm f$; the positive half has the same shape as the one-sided picture, with every height halved.

4. Converting rad/s to Hz

The messages are given with ω in rad/s. The frequency in hertz is $f = \omega/2\pi$; for example 377 rad/s is $377/2\pi = 60.0$ Hz.

2. Answer

Assumption: the modulation is DSB-SC with a unit carrier, $\varphi(t) = m(t)\cos(2\pi \times 10^5 t)$, as in Lathi's DSB-SC examples; $f_c = 100$ kHz.

Step 1: Signal (i), $m(t) = 1 + \cos 377t$

The message has a dc part 1 and a tone at $f_m = 377/2\pi = 60$ Hz.

$$\varphi(t) = \cos\omega_c t + \cos 377t\cos\omega_c t = \underbrace{\cos\omega_c t}_{\text{carrier, 100 kHz}} + \underbrace{\tfrac{1}{2}\cos(\omega_c - 377)t}_{\text{LSB, 99.94 kHz}} + \underbrace{\tfrac{1}{2}\cos(\omega_c + 377)t}_{\text{USB, 100.06 kHz}}$$

Step 2: Signal (ii), $m(t) = 2\sin^2(100\pi t)$

First write it as a sum of cosines with $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$, $x = 100\pi t$:

$$m(t) = 2 \cdot \tfrac{1}{2}\left[1 - \cos(200\pi t)\right] = 1 - \cos(200\pi t)$$

So it has a dc part 1 and a tone at $f_m = 200\pi/2\pi = 100$ Hz with amplitude -1 (a cosine with a phase of π).

$$\varphi(t) = \underbrace{\cos\omega_c t}_{\text{carrier, 100 kHz}} - \underbrace{\tfrac{1}{2}\cos(\omega_c - 200\pi)t}_{\text{LSB, 99.9 kHz}} - \underbrace{\tfrac{1}{2}\cos(\omega_c + 200\pi)t}_{\text{USB, 100.1 kHz}}$$

Step 3: Signal (iii), $m(t) = 3\cos 1000t\cos 3000t$

First turn the product into a sum, with $A = 3000t$, $B = 1000t$:

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$$m(t) = \tfrac{3}{2}[\cos 2000t + \cos 4000t]$$

The message has two tones, at $2000/2\pi = 318.31$ Hz and $4000/2\pi = 636.62$ Hz, each of amplitude 1.5, and no dc part. Each tone splits into two lines of amplitude 0.75:

$$\varphi(t) = 0.75\cos(\omega_c \pm 2000)t + 0.75\cos(\omega_c \pm 4000)t$$

with lines at 100 ± 0.31831 kHz and 100 ± 0.63662 kHz. There is no line at 100 kHz, because the message has no dc part.

Step 4: The spectra

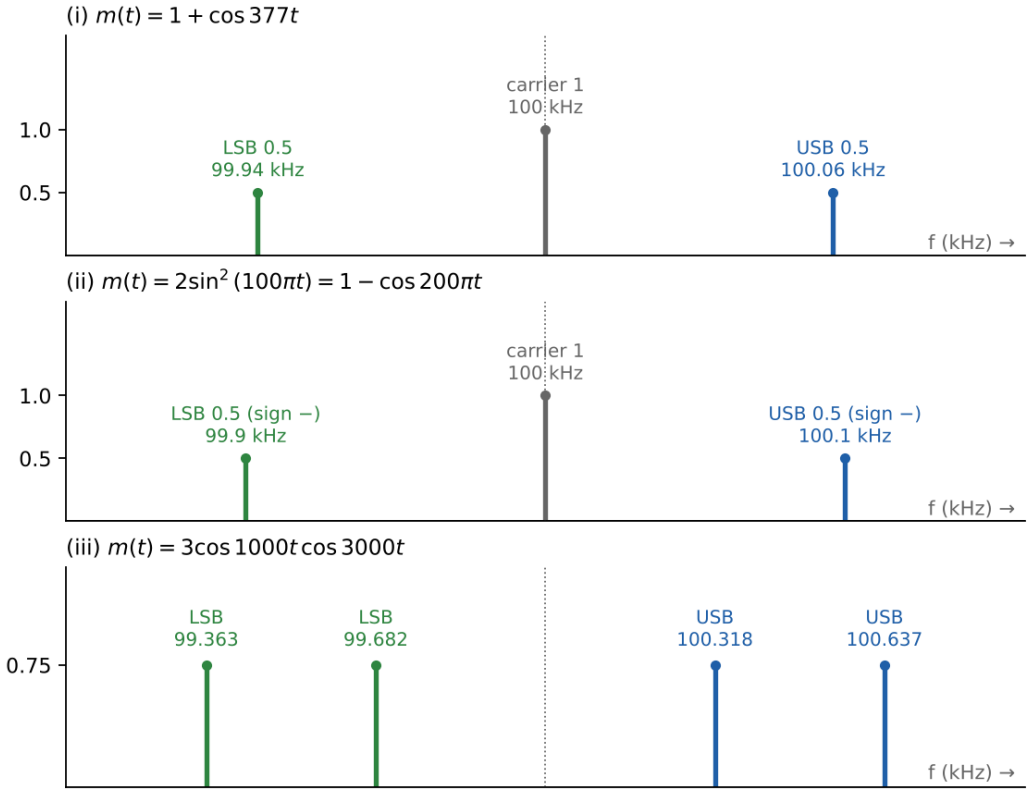


Figure 3: One-sided amplitude spectra around the 100 kHz carrier: green lines are the LSB, blue lines the USB, grey the carrier line

Reading the figure. Each panel is centred on 100 kHz (dotted line). In (i) and (ii) the grey line of height 1 at 100 kHz comes from the dc part of the message; the green LSB line and the blue USB line, each 0.5 high, sit 60 Hz and 100 Hz away. In (ii) both sideband lines carry a minus sign (phase π), which a magnitude plot shows as height 0.5. In (iii) there is no carrier line, and two pairs of lines of height 0.75 sit 318 Hz and 637 Hz away from the carrier.

Signal	Message frequencies	LSB (amplitude)	Carrier line	USB (amplitude)
(i)	dc, 60 Hz	99.94 kHz (0.5)	100 kHz (1)	100.06 kHz (0.5)
(ii)	dc, 100 Hz	99.9 kHz (−0.5)	100 kHz (1)	100.1 kHz (−0.5)
(iii)	318.31 Hz, 636.62 Hz	99.363, 99.682 kHz (0.75 each)	none	100.318, 100.637 kHz (0.75 each)

Table 1: Summary

Final Answer: see the table; the USB holds the lines above 100 kHz, the LSB the lines below, each sideband line has half the message tone's amplitude, and a dc part of the message appears as a line at 100 kHz.

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3. Keyword Explanation

1. Sideband

↩ back

The group of spectral lines on one side of the carrier. The USB is a copy of the message spectrum shifted up by f_c; the LSB is its mirror image below f_c.

4. Notes and Remarks

- In the two-sided (Lathi) spectrum each height above is halved and repeated at negative frequencies: for (i), $\frac{1}{2}$ at ± 100 kHz and $\frac{1}{4}$ at ± 99.94 and ± 100.06 kHz.
- If the course reads the question as conventional AM, $[A + m(t)]\cos\omega_c t$, only the carrier line grows by A; the sideband frequencies and amplitudes stay as found here.

1.(c) States Shannon's capacity theorem. A user wishes to transmit data over a channel having bandwidth B = 3000 Hz. If the SNR is 9.08 find the theoretical maximum bit rate of the channel. Also find the required SNR if the desired bitrate of the channel is doubled. Marks: 6

1. Theory

1. What limits the bit rate of a channel? (Core Idea)

Two things: the **bandwidth** B (how fast the signal can change) and the **noise** (how finely the receiver can tell levels apart). With more bandwidth we can send more symbols per second; with less noise we can pack more bits into each symbol. Shannon combined both in one formula.

2. Shannon's capacity theorem

For a channel of bandwidth B hertz with additive white Gaussian noise and signal-to-noise power ratio $\mathrm{SNR} = S/N$, the **capacity** is

$$C = B\log_2(1 + \mathrm{SNR}) \quad \text{bits/s}$$

The theorem: if the information rate R is at most C, there is a coding method that makes the error probability as small as we like; if $R > C$, no method can. So C is the theoretical maximum bit rate of the channel. It does not tell us how to build the code; it gives the limit.

Why a logarithm? $1 + \mathrm{SNR}$ is roughly the number of distinguishable levels squared: the number of levels grows like $\sqrt{1 + \mathrm{SNR}}$, and each symbol carries $\log_2$ of the number of levels; with $2B$ symbols per second (Nyquist) the rate is $2B\log_2\sqrt{1 + \mathrm{SNR}} = B\log_2(1 + \mathrm{SNR})$.

3. The SNR is a ratio, not decibels

In the formula, SNR is the plain power ratio. If it is given in decibels, first convert it: $\mathrm{SNR} = 10^{\mathrm{SNR}_{dB}/10}$.

2. Answer

Shannon's capacity theorem: a channel of bandwidth B and signal-to-noise ratio S/N can carry information without error (with suitable coding) at any rate up to

$$C = B\log_2\left(1 + \frac{S}{N}\right)$$

and at no higher rate.

Given:

- $B = 3000$ Hz
- $\mathrm{SNR} = 9.08$ (a power ratio)

Step 1: Maximum bit rate

$$C = 3000\log_2(1 + 9.08) = 3000\log_2(10.08) = 3000 \times 3.33342 = 10\,000.3 \text{ bits/s}$$

$$\boxed{C \approx 10 \text{ kbps}}$$

Step 2: SNR needed for double the bit rate

Set $C' = 2C = 20\,000.5$ bits/s and solve Shannon's formula for the SNR:

$$\log_2(1 + \text{SNR}') = \frac{C'}{B} = \frac{20\,000.5}{3000} = 6.66685$$

$$\text{SNR}' = 2^{6.66685} - 1 = 101.606 - 1 = 100.6$$

$$\boxed{\text{SNR}' \approx 100.6 \ (20.03 \text{ dB})}$$

A quicker way to see it: doubling C with the same B doubles $\log_2(1+\text{SNR})$, so $1+\text{SNR}' = (1+\text{SNR})^2 = 10.08^2 = 101.61$.

Final Answer: $C \approx 10$ kbps; to double it to 20 kbps the SNR must rise from 9.08 (9.58 dB) to about 100.6 (20.03 dB), more than ten times the signal power.

3. Notes and Remarks

- **Other reading:** if "SNR is 9.08" meant 9.08 dB, then SNR $= 10^{0.908} = 8.091$ and $C = 3000 \log_2 9.091 = 9553$ bits/s. The numbers suggest the ratio was meant, since it gives exactly 10 kbps.
- Doubling the rate costs far more than double the power, because capacity grows only with the logarithm of SNR. Doubling the bandwidth instead would double the capacity directly (if the noise did not grow with it).

1.(d) Find the equivalent noise resistance for several amplifiers in cascade. Marks: 7

1. Theory

1. Every resistor makes noise (Core Idea)

The electrons in a resistor move randomly because of heat. This makes a small random voltage across it, called **thermal (Johnson) noise**. Its mean-square value over a bandwidth B is

$$\overline{v_n^2} = 4kTBR$$

where $k = 1.38 \times 10^{-23}$ J/K is Boltzmann's constant, T the temperature in kelvin and R the resistance. The noise is proportional to R: a bigger resistor makes more noise.

2. Noise resistance of an amplifier

An amplifier's own transistors and resistors also make noise. Kennedy and Davis describe it with a trick: imagine the amplifier is **noiseless**, and put one extra imaginary resistor at its input that makes the same noise at the output. That resistor is the amplifier's **equivalent noise resistance** R_{eq}. Once a stage is described this way, noise can be compared and added using resistances only.

3. Why the gains are squared

A noise voltage v at the input of a stage of voltage gain A becomes Av at its output, so its mean-square value (its power) becomes $A^2\overline{v^2}$. Noise powers from separate sources are uncorrelated, so they **add** (their voltages do not).

2. Answer

Below, Step 1 is marked **Basic**: it recalls the noise formula the derivation stands on; a student who knows it can skip it.

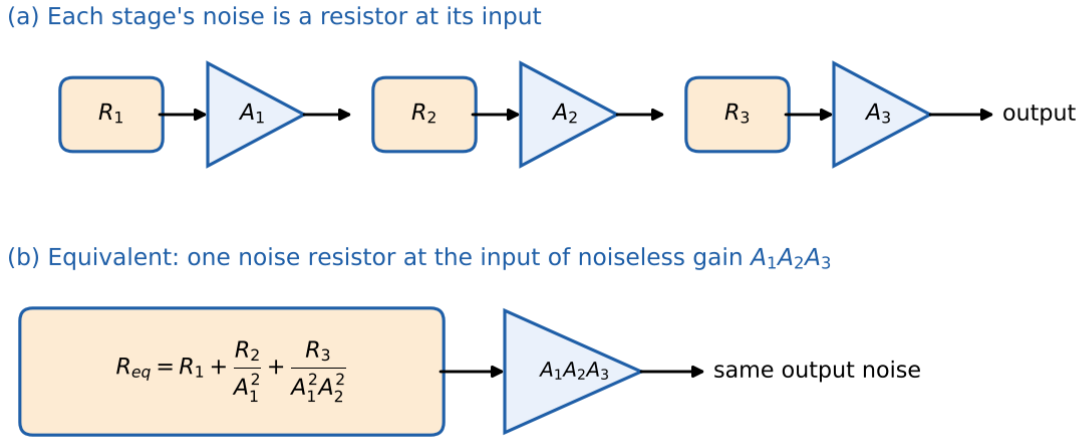


Figure 4: Amplifiers in cascade (a) and the single equivalent noise resistor at the input (b)

Reading the figure. In (a) each amplifier stage is drawn as a noiseless triangle with gain A_1, A_2, A_3, and the orange box in front of each is the resistor that stands for all the noise generated at that stage's input. In (b) the whole chain is replaced by one orange resistor R_{eq} in front of a single noiseless amplifier of gain $A_1A_2A_3$ that gives the same output noise.

Step 1 (Basic): Noise of a resistor

A resistor R at temperature T produces a noise voltage of mean-square value $\overline{v^2} = 4kTBR$ over a bandwidth B.

Step 2: Model each stage

Let the cascade have stages of voltage gain $A_1, A_2, A_3, \ldots$. All the noise generated at the input of stage i (its own noise plus that of the resistors connected there) is represented by a resistance R_i in front of a noiseless amplifier:

- R_1: the source (input) resistance plus the first stage's own equivalent noise resistance,
- R_2: the output resistance of stage 1 in parallel with the input resistance of stage 2, plus stage 2's equivalent noise resistance,
- R_3: the same for stage 3, and so on.

Step 3: Output noise of the cascade

Each noise voltage is amplified by every stage after it. Taking three stages:

- R_1 is amplified by $A_1A_2A_3$: output noise power $4kTBR_1\,(A_1A_2A_3)^2$
- R_2 is amplified by A_2A_3: output noise power $4kTBR_2\,(A_2A_3)^2$
- R_3 is amplified by A_3: output noise power $4kTBR_3\,A_3^2$

The three sources are independent, so their powers add:

$$\overline{v_{no}^2} = 4kTB\left[R_1A_1^2A_2^2A_3^2 + R_2A_2^2A_3^2 + R_3A_3^2\right]$$

Step 4: Refer the noise to the input

We want a single resistor R_{eq} at the input of a noiseless chain of gain $A_1A_2A_3$ that produces the same output noise:

$$4kTB\,R_{eq}\,(A_1A_2A_3)^2 = 4kTB\left[R_1A_1^2A_2^2A_3^2 + R_2A_2^2A_3^2 + R_3A_3^2\right]$$

Cancel $4kTB$ and divide both sides by $(A_1A_2A_3)^2$:

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$$R_{eq} = R_1 + \frac{R_2 A_2^2 A_3^2}{A_1^2 A_2^2 A_3^2} + \frac{R_3 A_3^2}{A_1^2 A_2^2 A_3^2}$$

Step 5: The result

$$\boxed{R_{eq} = R_1 + \frac{R_2}{A_1^2} + \frac{R_3}{A_1^2 A_2^2} + \cdots + \frac{R_n}{A_1^2 A_2^2 \cdots A_{n-1}^2}}$$

where A_i are the **voltage** gains of the stages.

Interpretation: each later stage's noise resistance is divided by the square of all the gain in front of it, so it matters less and less. With a first-stage gain of 10, the second stage's noise counts only 1/100 as much. **The first stage decides the noise of the whole receiver**, so it should be a low-noise amplifier with high gain (the same idea as Friis' formula for noise figure).

Worked example (illustrative values)

Stage 1: gain 10, input resistor 600 Ω, own noise resistance 1600 Ω, output resistor 27 kΩ. Stage 2: gain 25, input resistor 81 kΩ, own noise resistance 10 kΩ, output resistor 1 MΩ. Then $R_1 = 600 + 1600 = 2200\ \Omega$; $R_2 = (27\,\mathrm{k} \parallel 81\,\mathrm{k}) + 10\,\mathrm{k} = 20.25\,\mathrm{k} + 10\,\mathrm{k} = 30.25\ \mathrm{k}\Omega$; $R_3 = 1\ \mathrm{M}\Omega$, and

$$R_{eq} = 2200 + \frac{30\,250}{10^2} + \frac{10^6}{10^2 \times 25^2} = 2200 + 302.5 + 16 = 2518.5\ \Omega$$

The first stage's 2200 Ω is 87 % of the total.

Steps to remember

Noise of a resistor $4kTBR$ → each stage = noiseless gain with R_i in front → output noise powers add, each multiplied by the squared gain after it → divide by the total squared gain → $R_{eq} = R_1 + R_2/A_1^2 + R_3/(A_1A_2)^2 + \ldots$

3. Keyword Explanation

1. Mean-square value

↩ back

The average of the square of a random voltage; it is proportional to the noise power, which is why noise is added in mean squares.

4. Notes and Remarks

- The same reasoning with noise factors gives Friis' formula, $F = F_1 + (F_2 - 1)/G_1 + (F_3 - 1)/(G_1G_2) + \ldots$, where the G_i are power gains ($G = A^2$ for equal resistances).
- The example's values follow the style of Kennedy and Davis's worked example; the paper gives no numbers.

5. Final Concise Answer

(a) Twisted pair: two insulated copper wires twisted so that noise cancels in their difference; UTP and STP, Cat 1 to 7; cheap, used for telephone and Ethernet; bandwidth and distance limited. Radio waves: 3 kHz to 1 GHz, omnidirectional, ground/sky/line-of-sight propagation, pass through walls, used for AM/FM radio and TV broadcasting.

(b) (i) carrier 1 at 100 kHz, LSB 0.5 at 99.94 kHz, USB 0.5 at 100.06 kHz; (ii) carrier 1 at 100 kHz, LSB −0.5 at 99.9 kHz, USB −0.5 at 100.1 kHz; (iii) no carrier, LSB 0.75 at 99.363 and 99.682 kHz, USB 0.75 at 100.318 and 100.637 kHz.

(c) $C = B\log_2(1 + \mathrm{SNR}) \approx 10$ kbps; doubling the rate needs SNR ≈ 100.6 (20 dB).

(d) $R_{eq} = R_1 + R_2/A_1^2 + R_3/(A_1^2A_2^2) + \ldots$; the first stage dominates.

2. (a) Show that the average information per message is equal to the entropy of the information source. Marks: 8

(b) An angle modulated signal with carrier frequency 10 MHz is described by the equation $s(t) = 12\cos(\omega_c t + \sin 3000\pi t + 10\sin 2000\pi t)$

(i) Find the power of the modulated signal
(ii) Find the frequency deviation Δf
(iii) Find the β
(iv) Find the bandwidth. Marks: 8

(c) Explain the effect of phase and frequency error during coherent detection of DSB-SC. Marks: 6

(d) Describe the condition on message signal spectrum for SSB generation using selective filtering. Marks: 3

2.(a) Show that the average information per message is equal to the entropy of the information source. Marks: 8

1. Theory

1. What is a source? (Core Idea)

A **source** produces messages one after another: letters from a keyboard, samples from a microphone. We model it as picking one message at a time from a fixed list $m_1, m_2, \ldots, m_n$, message m_i with probability P_i, where $\sum P_i = 1$. A **memoryless** source picks each message independently of the earlier ones.

2. How much information is in one message?

"The sun rose today" tells you nothing new; "it snowed in Dhaka" tells you a lot. Information grows when the probability falls. Hartley and Shannon measure it as

$$I_i = \log_2 \frac{1}{P_i} \quad \text{bits}$$

- A certain message ($P = 1$) carries 0 bits.
- A fair coin toss ($P = \frac{1}{2}$) carries 1 bit, which defines the unit.
- For two independent messages the probabilities multiply, $P = P_1 P_2$, and the logarithm turns the product into a sum, $I = I_1 + I_2$: information adds, as it should.

3. What is entropy?

Different messages carry different amounts of information. The **entropy** $H(m)$ of the source is defined as the probability-weighted average,

$$H(m) = \sum_{i=1}^{n} P_i \log_2 \frac{1}{P_i}$$

The question asks us to show that this definition is exactly what we get if we count the information in a long message and divide by the number of messages.

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2. Answer

Below, Step 1 is marked **Basic** (the information measure); a student who knows it can skip it.

Step 1 (Basic): Information in one message

A message of probability P_i carries $I_i = \log_2(1/P_i)$ bits.

Step 2: A long sequence of messages

Let the source emit a long sequence of N messages. Because the source is memoryless and N is large, the relative frequency of each message approaches its probability (law of large numbers), so message m_i appears about

$$N_i = NP_i \quad \text{times}$$

Step 3: Total information in the sequence

Each occurrence of m_i brings I_i bits, and information from independent messages adds. So the total information is

$$I_{total} = \sum_{i=1}^{n} N_i I_i = \sum_{i=1}^{n} NP_i \log_2 \frac{1}{P_i}$$

Step 4: Average per message

Divide the total by the number of messages N:

$$\frac{I_{total}}{N} = \sum_{i=1}^{n} P_i \log_2 \frac{1}{P_i}$$

Step 5: Compare with the entropy

The right-hand side is exactly the entropy $H(m)$. Therefore

$$\boxed{\text{average information per message} = \frac{I_{total}}{N} = \sum_{i=1}^{n} P_i \log_2 \frac{1}{P_i} = H(m) \quad \text{bits/message}}$$

which is what we had to show (Lathi, Sec. 13.1).

Check with an example. A source has four messages with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}$. In $N = 800$ messages there are about 400, 200, 100 and 100 of each, carrying 1, 2, 3 and 3 bits. Total $= 400 + 400 + 300 + 300 = 1400$ bits, so the average is $1400/800 = 1.75$ bits; the entropy is $\frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{8}(3) = 1.75$ bits. They agree.

Key Points:

- $0 \le H(m) \le \log_2 n$: zero when one message is certain, the maximum $\log_2 n$ when all n messages are equally likely.
- If the source sends r messages per second, the **information rate** is $R = rH$ bits/s.

Steps to remember

$I_i = \log_2(1/P_i) \to$ in N messages, m_i occurs NP_i times $\to$ total $= \sum NP_iI_i \to$ divide by $N \to \sum P_i \log_2(1/P_i) = H$.

3. Keyword Explanation

1. Entropy

↩ back

The average information per message of a source, in bits per message; also the smallest average number of binary digits per message that any lossless code can reach (Shannon's source coding theorem).

4. Notes and Remarks

- The argument needs the sequence to be long, so that the counts N_i are close to NP_i; this is why entropy is a long-run average.

2.(b) An angle modulated signal with carrier frequency 10 MHz is described by the equation $s(t) = 12\cos(\omega_c t + \sin 3000\pi t + 10\sin 2000\pi t)$ (i) Find the power of the modulated signal (ii) Find the frequency deviation Δf (iii) Find the β (iv) Find the bandwidth. Marks: 8

1. Theory

1. What is angle modulation? (Core Idea)

In angle modulation the amplitude of the carrier stays constant, and the message goes into its angle:

$$s(t) = A\cos\theta(t), \qquad \theta(t) = \omega_c t + \phi(t)$$

where $\phi(t)$ carries the message.

2. Instantaneous frequency

The rate of change of the angle is the **instantaneous frequency**:

$$\omega_i(t) = \frac{d\theta}{dt} = \omega_c + \frac{d\phi}{dt}, \qquad f_i(t) = f_c + \frac{1}{2\pi}\frac{d\phi}{dt}$$

The largest distance of f_i from f_c is the **frequency deviation** Δf.

3. Power

The amplitude never changes, so the power is that of a plain sinusoid: $P = A^2/2$ (into 1 Ω), whatever the message.

4. Deviation ratio and bandwidth

With B the highest frequency in the message, the **deviation ratio** is $\beta = \Delta f/B$, and Carson's rule estimates the bandwidth as $B_T = 2(\Delta f + B) = 2B(\beta + 1)$.

2. Answer

Given: $s(t) = 12\cos(\omega_c t + \sin 3000\pi t + 10\sin 2000\pi t)$, $f_c = 10$ MHz.

Step 1: Power

$$P = \frac{A^2}{2} = \frac{12^2}{2} = 72 \text{ W}$$

$$\boxed{P = 72 \text{ W (normalized to 1 } \Omega)}$$

Into a load R_L the power is $72/R_L$ W.

Step 2: Instantaneous frequency

The angle is $\theta(t) = \omega_c t + \sin 3000\pi t + 10\sin 2000\pi t$. Differentiate, using $\frac{d}{dt}\sin at = a\cos at$:

$$\omega_i(t) = \omega_c + 3000\pi\cos 3000\pi t + 20\,000\pi\cos 2000\pi t$$

Step 3: Frequency deviation

Both cosines reach +1 together at $t = 0$, so the largest deviation is the sum of the two amplitudes:

$$\Delta\omega = 3000\pi + 20\,000\pi = 23\,000\pi \text{ rad/s}$$

$$\Delta f = \frac{\Delta\omega}{2\pi} = \frac{23\,000\pi}{2\pi} = 11\,500 \text{ Hz}$$

$$\boxed{\Delta f = 11.5 \text{ kHz}}$$

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Step 4: Deviation ratio

The message tones are at $3000\pi/2\pi = 1500$ Hz and $2000\pi/2\pi = 1000$ Hz, so the message bandwidth is the highest, $B = 1500$ Hz:

$$\beta = \frac{\Delta f}{B} = \frac{11\,500}{1500} = 7.67$$

$$\boxed{\beta \approx 7.67}$$

Step 5: Bandwidth (Carson's rule)

$$B_T = 2(\Delta f + B) = 2(11\,500 + 1500) = 26\,000 \text{ Hz}$$

$$\boxed{B_T \approx 26 \text{ kHz}}$$

Final Answer: $P = 72$ W, $\Delta f = 11.5$ kHz, $\beta \approx 7.67$, $B_T \approx 26$ kHz.

3. Notes and Remarks

- The peak phase deviation is $\Delta\phi = 1 + 10 = 11$ rad.
- Lathi has a worked example of the same form with $5\sin 3000t$; this paper writes $\sin 3000\pi t$, and the answer follows the paper.
- β uses the highest message frequency (Lathi's definition $\beta = \Delta f/B$); with the 1000 Hz tone instead one would get 11.5, which is not the deviation ratio.

2.(c) Explain the effect of phase and frequency error during coherent detection of DSB-SC. Marks: 6

1. Theory

1. Coherent (synchronous) detection (Core Idea)

A DSB-SC signal $m(t)\cos\omega_c t$ has no carrier line, so an envelope detector cannot recover $m(t)$. The receiver instead multiplies the signal by a locally made carrier $\cos\omega_c t$ and low-pass filters it. This needs a local carrier with **exactly** the same frequency and phase as the transmitter's: that is what "coherent" means.

2. Why the local carrier is never perfect

The local oscillator is a separate circuit, far from the transmitter. Its frequency drifts a little with temperature, and the path delay shifts its phase. So in practice the local carrier is $\cos[(\omega_c + \Delta\omega)t + \delta]$, with a small frequency error $\Delta\omega$ and phase error δ.

2. Answer

Below, Step 2 is marked **Basic** (the product identity); a student who knows it can skip it.

Step 1: The received signal and the local carrier

$$\varphi(t) = m(t)\cos\omega_c t, \qquad c(t) = \cos[(\omega_c + \Delta\omega)t + \delta]$$

Step 2 (Basic): The product identity

$$\cos A \cos B = \tfrac{1}{2}[\cos(A - B) + \cos(A + B)]$$

Step 3: Multiply

With $A = (\omega_c + \Delta\omega)t + \delta$ and $B = \omega_c t$:

$$e(t) = m(t)\cos\omega_c t\cos[(\omega_c + \Delta\omega)t + \delta] = \tfrac{1}{2}m(t)\cos(\Delta\omega t + \delta) + \tfrac{1}{2}m(t)\cos[(2\omega_c + \Delta\omega)t + \delta]$$

Step 4: Low-pass filter

The second term sits around $2f_c$ and is removed by the LPF. The output is

$$\boxed{e_o(t) = \tfrac{1}{2} m(t) \cos(\Delta\omega t + \delta)}$$

Step 5: Phase error only ($\Delta\omega = 0$)

$$e_o(t) = \tfrac{1}{2} m(t) \cos\delta$$

- The output is the right shape but scaled by $\cos\delta$: **attenuation**, no distortion, if δ is constant.
- At $\delta = \pm 90°$ the output is *zero*: the **quadrature null effect**.
- If δ wanders randomly (for example with changing path delay), the output fades in and out.

Step 6: Frequency error only ($\delta = 0$)

$$e_o(t) = \tfrac{1}{2} m(t) \cos\Delta\omega t$$

- The message is multiplied by a slow sinusoid: its level **rises and falls** at the rate Δf (a beating or warbling effect). Even a few hertz of error is very audible, and it is serious distortion.
- Both errors are why DSB-SC needs carrier recovery (a pilot carrier, a squaring loop or a Costas loop).

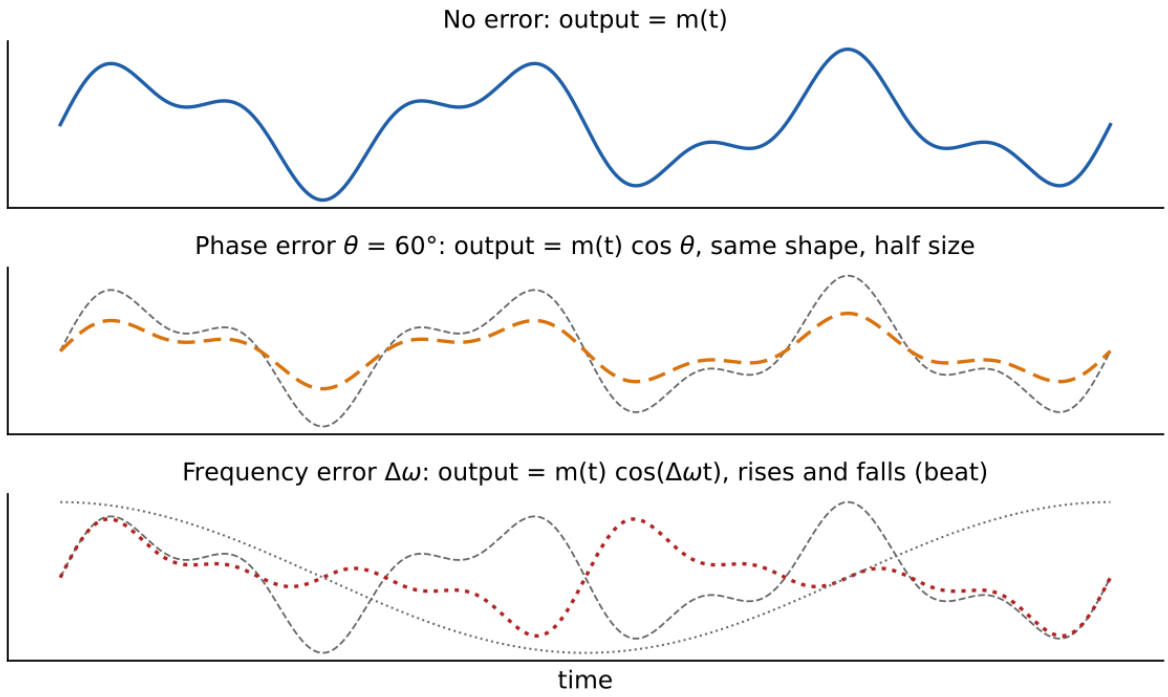


Figure 5: Output without error (top), with a 60° phase error (middle) and with a frequency error (bottom); reused from the 2021-22 paper

Reading the figure. The top panel is the correct output $m(t)$ (blue). In the middle the dashed orange output has the same shape at half the size, because $\cos 60° = 0.5$. In the bottom panel the dotted red output keeps rising and falling with the slow grey cosine $\cos\Delta\omega t$, so the message keeps fading in and out.
Final Summary: the output is $\frac{1}{2} m(t) \cos(\Delta\omega t + \delta)$; a phase error attenuates the output by $\cos\delta$ (zero at 90°), and a frequency error multiplies it by a slowly varying $\cos\Delta\omega t$, which causes beating and distortion.

3. Keyword Explanation

1. Quadrature null effect

↩ back

The complete loss of the output of a coherent detector when its local carrier is 90° out of phase with the received carrier.

4. Notes and Remarks

- Lathi (Sec. 4.2 and 4.6) gives the same result; a Costas receiver locks the local carrier to the incoming one so that $\Delta\omega$ and δ stay near zero.

2.(d) Describe the condition on message signal spectrum for SSB generation using selective filtering. Marks: 3

1. Theory

1. SSB by selective filtering (Core Idea)

The simplest way to make SSB is to make DSB-SC first and then remove one sideband with a **bandpass filter**. The two sidebands of a DSB-SC signal sit right next to each other, one above and one below f_c. A real filter cannot jump from "pass" to "stop" at one frequency; it needs a **transition band** to change gradually.

2. Answer

Condition: the message must have **little or no power near zero frequency** (a "hole" at dc), so that a gap separates the upper and lower sidebands around f_c in which the filter's response can fall from pass to stop.

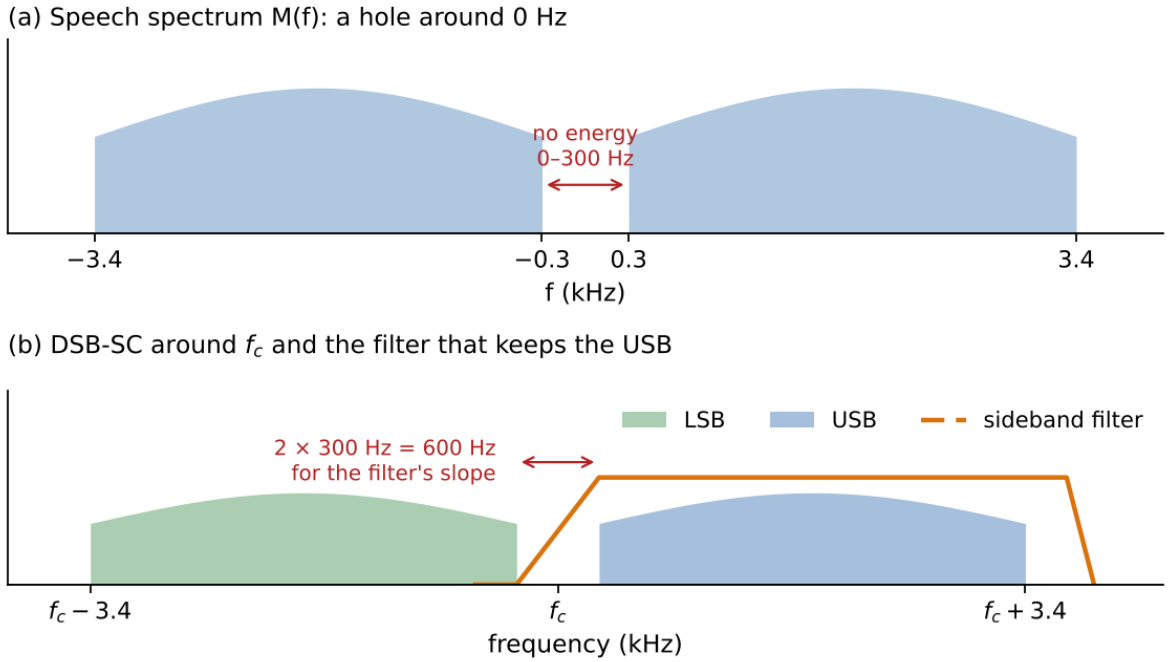


Figure 6: (a) Speech has no energy below about 300 Hz; (b) so its USB and LSB are 600 Hz apart, room for the filter's slope

Reading the figure. In (a) the blue speech spectrum is empty between −300 Hz and +300 Hz. In (b), after DSB-SC modulation, the green LSB and the blue USB are separated by a 600 Hz gap around f_c, and the orange sideband filter rises across that gap, passing the USB and stopping the LSB.

- If the message has no energy below f_L, the gap between the sidebands is $2f_L$, and the filter's transition band must fit in it. Speech ($f_L \approx 300$ Hz) gives a 600 Hz gap, so speech suits SSB by filtering.
- Signals with large power down to dc (television video, data with a dc part) have no gap, so the sidebands cannot be separated: they use **VSB** instead.
- The higher the carrier, the harder a 600 Hz transition becomes, so the filtering is often done at a low carrier first and the signal is then shifted up.

3. Notes and Remarks

- Lathi (Sec. 4.4) states the same condition: selective filtering works for signals with "little or no power near the origin".

4. Final Concise Answer

(a) In N messages m_i occurs NP_i times, carrying $\log_2(1/P_i)$ bits each; the total divided by N is $\sum P_i \log_2(1/P_i) = H$.

(b) $P = 72$ W (1 Ω), $\Delta f = 11.5$ kHz, $\beta = 7.67$, $B_T = 26$ kHz.

(c) Output $\frac{1}{2}m(t)\cos(\Delta\omega t + \delta)$: a phase error attenuates by $\cos\delta$ (null at 90°); a frequency error causes beating.

(d) The message must have little or no power near dc, leaving a gap $2f_L$ between the sidebands for the filter's transition band.

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3. (a) Using proper block diagram explain phase shifting method of SSB generation. Marks: 8
   (b) Explain the effect of variation of amplitude and frequency of message signal on the spectrum and bandwidth of FM. Marks: 8
   (c) Design an Armstrong indirect FM modulator to generate an FM signal with carrier frequency 97.3 MHz and $\Delta f = 10.24\ kHz$. A NBFM generator of $f_{c1} = 20\ kHz$ and $\Delta f = 5\ Hz$ is available. Only frequency doublers can be used as multipliers. Additionally, a local oscillator (LO) with adjustable frequency between 400 and 500 kHz is readily available for frequency mixing. Marks: 9

3.(a) Using proper block diagram explain phase shifting method of SSB generation. Marks: 8

1. Theory

1. What is SSB? (Core Idea)

A DSB-SC signal has two sidebands that carry the same information (one is the mirror image of the other). **Single sideband (SSB)** sends only one of them, so it needs only the message bandwidth B instead of $2B$, and no power is wasted on a carrier or a duplicate sideband.

2. The Hilbert transform: a −90° phase shifter

The phase-shift method needs a circuit that shifts the phase of *every* frequency component of $m(t)$ by $-90°$ $(-\pi/2)$ without changing its amplitude. Its output is called the **Hilbert transform** of $m(t)$, written $m_h(t)$. Its transfer function is

$$H(f) = -j\,\mathrm{sgn}(f) = \begin{cases} -j = e^{-j\pi/2}, & f > 0 \\ +j = e^{+j\pi/2}, & f < 0 \end{cases}$$

For a tone this simply turns a cosine into a sine: if $m(t) = \cos\omega_m t$ then $m_h(t) = \cos(\omega_m t - \pi/2) = \sin\omega_m t$.

2. Answer

The phase-shift method builds the SSB signal from its time-domain formula (Lathi, Sec. 4.4):

$$\varphi_{SSB}(t) = m(t)\cos\omega_c t \mp m_h(t)\sin\omega_c t \qquad (- : \text{USB},\ + : \text{LSB})$$

Steps 2 and 3 below are marked **Basic** (the tone case and the identity); a student who knows them can skip them.

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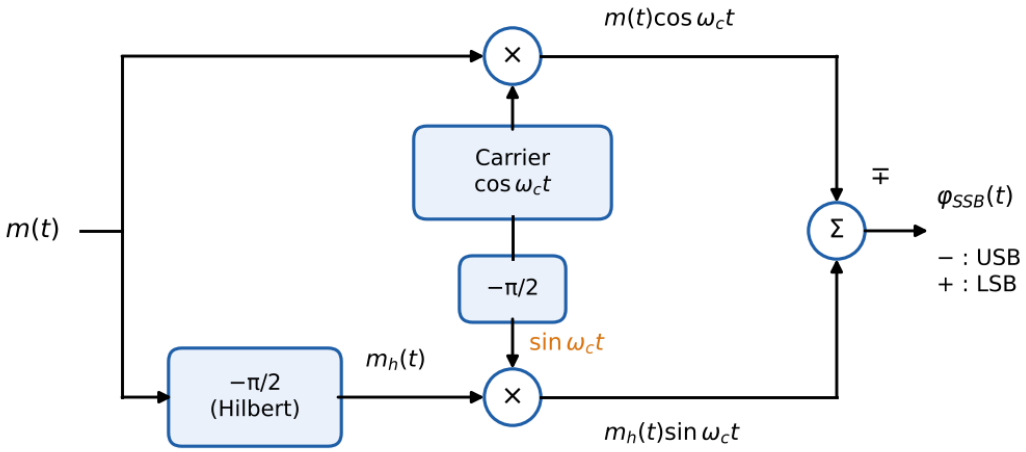


Figure 7: Phase-shift SSB modulator: two DSB-SC branches, one fed through a $-90°$ shifter, added or subtracted

Reading the figure. The message $m(t)$ splits into two paths. The upper path multiplies it by the carrier $\cos\omega_c t$. The lower path first shifts every component of $m(t)$ by $-90°$ (Hilbert transformer, giving $m_h(t)$) and then multiplies it by the carrier shifted by $-90°$, which is $\sin\omega_c t$ (orange). The summer subtracts the lower product for the USB or adds it for the LSB.

Step 1: The block diagram

- **Upper branch:** a balanced modulator forms $m(t)\cos\omega_c t$ (DSB-SC).
- **Lower branch:** a wideband $-90°$ phase shifter forms $m_h(t)$, and a second balanced modulator multiplies it by $\sin\omega_c t$, obtained by shifting the carrier by $-90°$.
- **Summer:** forms $m(t)\cos\omega_c t - m_h(t)\sin\omega_c t$ (USB) or $m(t)\cos\omega_c t + m_h(t)\sin\omega_c t$ (LSB).

Step 2 (Basic): A tone through the two branches

Let $m(t) = \cos\omega_m t$. The $-90°$ shifter gives $m_h(t) = \sin\omega_m t$. So the two branch outputs are $\cos\omega_m t\cos\omega_c t$ and $\sin\omega_m t\sin\omega_c t$.

Step 3 (Basic): The identities

$$\cos A\cos B = \tfrac{1}{2}[\cos(A-B)+\cos(A+B)], \qquad \sin A\sin B = \tfrac{1}{2}[\cos(A-B)-\cos(A+B)]$$

with $A = \omega_c t$, $B = \omega_m t$:

$$\cos\omega_m t\cos\omega_c t = \tfrac{1}{2}\cos(\omega_c-\omega_m)t + \tfrac{1}{2}\cos(\omega_c+\omega_m)t$$

$$\sin\omega_m t\sin\omega_c t = \tfrac{1}{2}\cos(\omega_c-\omega_m)t - \tfrac{1}{2}\cos(\omega_c+\omega_m)t$$

Step 4: Subtract for the USB, add for the LSB

$$\text{difference} = \cos(\omega_c+\omega_m)t \quad \text{(only the upper side frequency)}$$

$$\text{sum} = \cos(\omega_c-\omega_m)t \quad \text{(only the lower side frequency)}$$

Each branch alone is a DSB-SC signal; the two branches have the same lower sideband and opposite upper sidebands (or the reverse), so one sideband cancels and the other doubles.

Step 5: Any message

A general $m(t)$ is a sum of tones (Fourier), and both branches are linear in m, so the cancellation happens for every component at once:

$$\boxed{\varphi_{USB}(t) = m(t)\cos\omega_c t - m_h(t)\sin\omega_c t, \qquad \varphi_{LSB}(t) = m(t)\cos\omega_c t + m_h(t)\sin\omega_c t}$$

Advantages:

- No sharp sideband filter is needed, so it works even for messages with energy near dc, and at any carrier frequency.
- The sideband can be switched by changing one sign.

Disadvantages:

- An ideal wideband $-90°$ shifter (a Hilbert transformer) is hard to build: every frequency of the message must be shifted by exactly 90°.
- The two branches must be balanced in gain and phase; any error lets part of the unwanted sideband through.

Steps to remember

Two branches $m\cos\omega_c t$ and $m_h \sin\omega_c t \to$ tone: $\cos\omega_m t$ becomes $\sin\omega_m t \to$ product identities $\to$ subtract: USB only, add: LSB only $\to$ any $m(t)$ by linearity.

3. Keyword Explanation

1. Balanced modulator

↩ back

A circuit whose output is the product of its two inputs without the carrier (a DSB-SC modulator).

2. Hilbert transformer

↩ back

An all-pass filter that shifts every positive frequency by $-90°$ and every negative frequency by $+90°$.

4. Notes and Remarks

- Weaver's method (a third SSB method) avoids the wideband phase shifter by using two stages of quadrature mixing with low-pass filters.

3.(b) Explain the effect of variation of amplitude and frequency of message signal on the spectrum and bandwidth of FM. Marks: 8

1. Theory

1. FM in one formula (Core Idea)

In FM the instantaneous frequency follows the message: $f_i(t) = f_c + k_f m(t)$, where k_f (Hz per volt) is the modulator's sensitivity. For a tone $m(t) = A_m \cos\omega_m t$:

$$\Delta f = k_f A_m, \qquad \beta = \frac{\Delta f}{f_m} = \frac{k_f A_m}{f_m}$$

So the **deviation** depends only on the message **amplitude**, and the **modulation index** depends on the ratio of amplitude to frequency.

2. The FM spectrum

A tone-modulated FM signal is

$$\varphi_{FM}(t) = A\cos(\omega_c t + \beta \sin\omega_m t) = A\sum_{n=-\infty}^{\infty} J_n(\beta)\cos(\omega_c + n\omega_m)t$$

It has a carrier line and an infinite set of sideband lines at $f_c \pm n f_m$, **spaced f_m apart**, with amplitudes $AJ_n(\beta)$ (Bessel functions). Only the lines with $n \le \beta + 1$ are significant, so the bandwidth is about

$$B_T \approx 2(\beta + 1) f_m = 2(\Delta f + f_m) \qquad \text{(Carson's rule)}$$

2. Answer

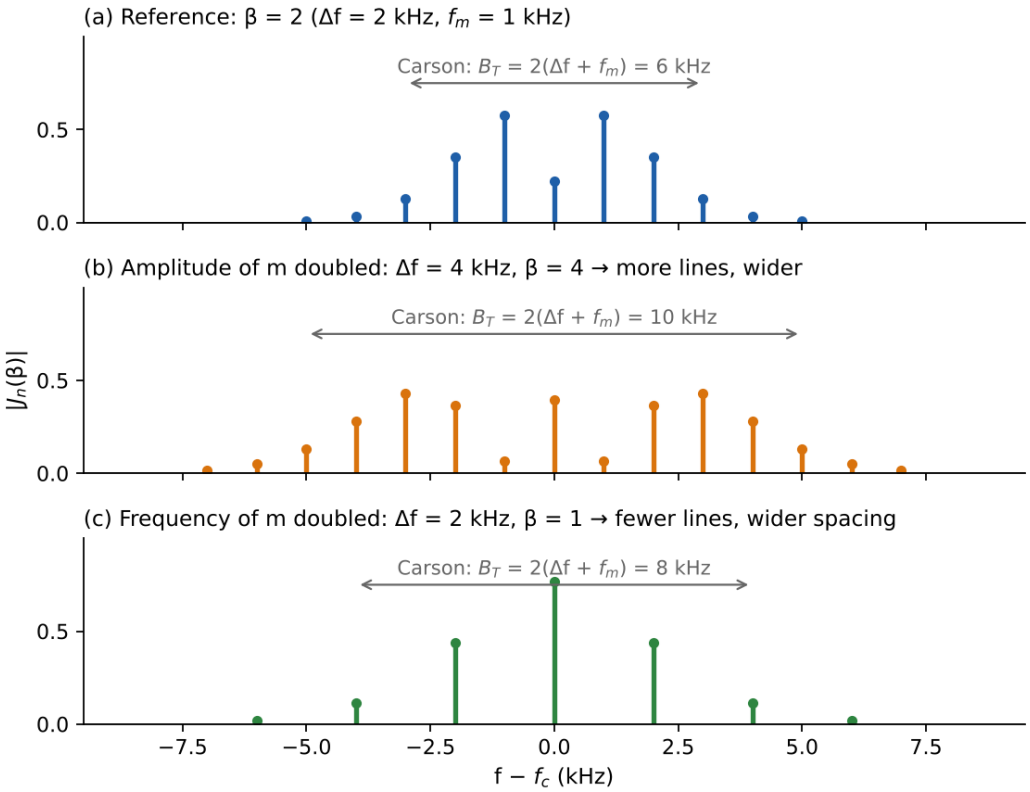


Figure 8: FM line spectra |J_n(β)|: (a) reference, (b) message amplitude doubled, (c) message frequency doubled; the grey arrow is Carson's bandwidth

Reading the figure. Each panel shows the line heights $|J_n(\beta)|$ against $f - f_c$. Panel (a), blue, is the reference with $\Delta f = 2$ kHz and $f_m = 1$ kHz ($\beta = 2$): lines 1 kHz apart, Carson bandwidth 6 kHz. In (b), orange, the amplitude is doubled: lines stay 1 kHz apart, but more of them are significant, and the bandwidth grows to 10 kHz. In (c), green, the frequency is doubled: the lines are 2 kHz apart, there are fewer significant ones ($\beta = 1$), and the bandwidth grows only to 8 kHz.

Effect of the message amplitude A_m (frequency fixed)

- $\Delta f = k_f A_m$ grows in proportion to A_m, and so does $\beta = \Delta f / f_m$.
- **Spectrum:** the line spacing stays f_m, but the energy spreads into *more* significant sidebands; the carrier amplitude $J_0(\beta)$ changes (it even vanishes at $\beta = 2.405$). The total power $A^2/2$ does not change; it is only redistributed.
- **Bandwidth:** $B_T = 2(k_f A_m + f_m)$ rises almost in proportion to A_m for wideband FM. This is the main thing that sets an FM signal's bandwidth.

Effect of the message frequency f_m (amplitude fixed)

- $\Delta f = k_f A_m$ does *not* change; $\beta = \Delta f / f_m$ falls as f_m rises.
- **Spectrum:** the lines move further apart (spacing f_m), but fewer of them are significant because β is smaller.
- **Bandwidth:** $B_T = 2(\Delta f + f_m)$ rises only a little: for wideband FM ($\Delta f \gg f_m$) it stays close to $2\Delta f$. Only for narrowband FM ($\beta \ll 1$, $B_T \approx 2f_m$) does it grow in proportion to f_m.

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Change	Δf	β	Line spacing	Number of significant lines	Bandwidth $2(\Delta f + f_m)$
A_m doubled	doubles	doubles	same	more	nearly doubles (6 → 10 kHz)
f_m doubled	same	halves	doubles	fewer	small rise (6 → 8 kHz)

Table 2: Summary

Final Summary: in FM the amplitude of the message sets the deviation and so mainly the bandwidth; the frequency of the message sets the spacing of the sidebands but changes the bandwidth only slightly. FM bandwidth is therefore nearly independent of the message frequency, which is why FM is called a constant-bandwidth system.

3. Keyword Explanation

1. Bessel function $J_n(\beta)$

↩ back

The amplitude of the n-th sideband of tone-modulated FM, relative to the unmodulated carrier.

4. Notes and Remarks

- In **PM** the opposite holds for frequency: $\Delta f = k_p A_m f_m$ grows with f_m, so a PM signal's bandwidth grows in proportion to the message frequency.

3.(c) Design an Armstrong indirect FM modulator to generate an FM signal with carrier frequency 97.3 MHz and $\Delta f = 10.24\ kHz$. A NBFM generator of $f_{c1} = 20\ kHz$ and $\Delta f = 5\ Hz$ is available. Only frequency doublers can be used as multipliers. Additionally, a local oscillator (LO) with adjustable frequency between 400 and 500 kHz is readily available for frequency mixing. Marks: 9

1. Theory

1. Why Armstrong's method? (Core Idea)

A crystal oscillator gives a very stable carrier but cannot be frequency-modulated much. Armstrong's indirect method therefore starts with **narrowband FM** (NBFM) made from a stable crystal oscillator (deviation only a few hertz), and then raises the deviation to the wanted value with **frequency multipliers**.

2. What a multiplier does

A frequency multiplier by n turns $\cos(\omega_c t + \phi(t))$ into $\cos(n\omega_c t + n\phi(t))$: it multiplies *both* the carrier frequency and the deviation by n. A doubler is $n = 2$; k doublers in cascade multiply by 2^k.

3. What a mixer does

A mixer multiplies the FM signal by a local oscillator (LO) and a bandpass filter keeps the sum or the difference frequency. It **moves the carrier** to $f_c \pm f_{LO}$ but leaves the **deviation unchanged**. So the mixer is used to correct the carrier frequency, which the multipliers have made too large, without spoiling the deviation.

2. Answer

Given: NBFM: $f_{c1} = 20$ kHz, $\Delta f_1 = 5$ Hz; output: $f_c = 97.3$ MHz, $\Delta f = 10.24$ kHz; only doublers; LO adjustable from 400 to 500 kHz.

Step 1: Total multiplication needed

The deviation is changed only by the multipliers, so their total factor must be

$$n_1 n_2 = \frac{\Delta f}{\Delta f_1} = \frac{10\,240}{5} = 2048 = 2^{11}$$

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So **11 doublers** in all. We split them as $n_1 = 2^a$ before the mixer and $n_2 = 2^{11-a}$ after it.

Step 2: Carrier frequencies around the mixer

- After the first multiplier: $f_2 = n_1 \times 20$ kHz
- Just before the second multiplier, the carrier must be $f_3 = \dfrac{97.3 \text{ MHz}}{n_2}$
- The mixer must turn f_2 into f_3: $f_3 = f_2 + f_{LO}$ (sum) or $f_3 = |f_2 - f_{LO}|$ (difference), with $400 \le f_{LO} \le 500$ kHz.

Step 3: Find the split

$f_3 - f_2 = \dfrac{97.3 \times 10^6\, n_1}{2048} - 20\,000\, n_1 = 47\,509.77\, n_1 - 20\,000\, n_1 = 27\,509.77\, n_1$ Hz. This must lie between 400 and 500 kHz:

$$14.54 \le n_1 \le 18.18 \quad \Rightarrow \quad n_1 = 16 = 2^4 \text{ (the only power of two)}$$

(Trying the difference $f_3 = f_{LO} - f_2$ needs $f_{LO} = 47\,509.77\, n_1$, between 400 and 500 kHz only for $8.4 \le n_1 \le 10.5$, which holds no power of two.)
So $n_1 = 16$ (4 doublers) and $n_2 = 2048/16 = 128$ (7 doublers).

Step 4: The frequencies

- After $\times 16$: $f_2 = 16 \times 20 = 320$ kHz, $\Delta f_2 = 16 \times 5 = 80$ Hz
- Needed after the mixer: $f_3 = 97.3$ MHz$/128 = 760.15625$ kHz, $\Delta f_3 = 80$ Hz (the mixer does not change it)
- LO: $f_{LO} = f_3 - f_2 = 760.15625 - 320 = 440.15625$ kHz, within 400 to 500 kHz; the BPF after the mixer keeps the **sum** frequency.
- After $\times 128$: $f_c = 128 \times 760.15625$ kHz $= 97.3$ MHz, $\Delta f = 128 \times 80 = 10\,240$ Hz $= 10.24$ kHz ✔

$$\boxed{n_1 = 16 \text{ (4 doublers)}, \quad f_{LO} = 440.15625 \text{ kHz (sum)}, \quad n_2 = 128 \text{ (7 doublers)}}$$

Step 5: The block diagram

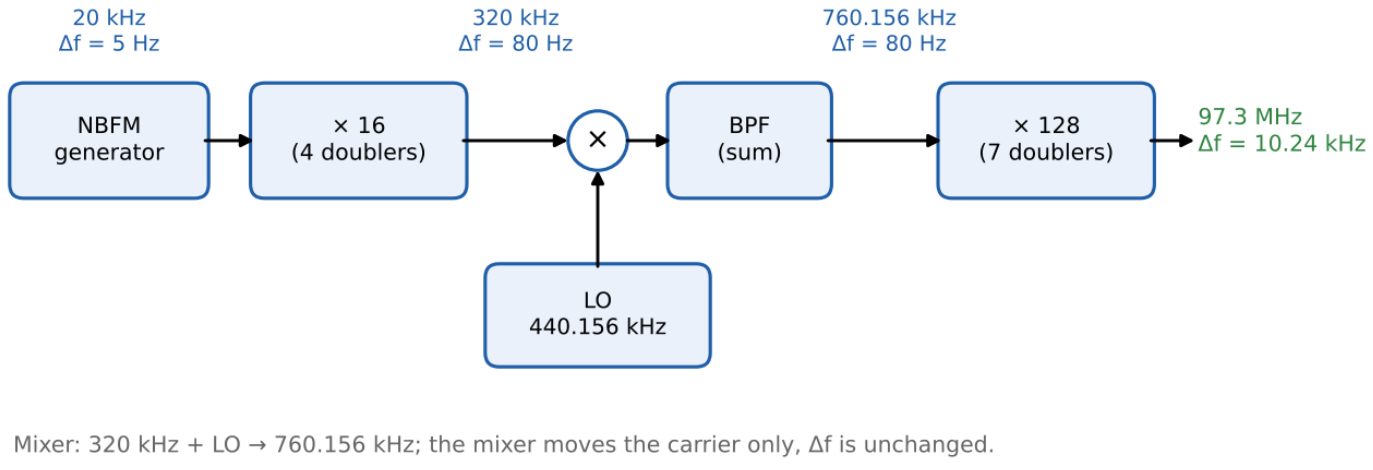


Figure 9: Armstrong indirect FM modulator for 97.3 MHz: frequencies and deviations at each point

Reading the figure. From left to right: the NBFM generator gives 20 kHz with 5 Hz deviation; the $\times 16$ multiplier (4 doublers) gives 320 kHz and 80 Hz; the mixer with the 440.156 kHz LO and the bandpass filter keep the sum, 760.156 kHz, still 80 Hz; the $\times 128$ multiplier (7 doublers) gives the final 97.3 MHz with 10.24 kHz deviation (green).

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Final Answer: 4 doublers ($\times 16$) $\rightarrow$ mixer with $f_{LO} = 440.15625$ kHz, keeping the sum 760.15625 kHz $\rightarrow$ 7 doublers ($\times 128$); output 97.3 MHz with $\Delta f = 10.24$ kHz.

Steps to remember

$n_1 n_2 = \Delta f / \Delta f_1 \rightarrow f_2 = n_1 f_{c1}$, $f_3 = f_c / n_2 \rightarrow f_{LO} = |f_3 - f_2|$ must fall in the LO range $\rightarrow$ pick the power of two that fits $\rightarrow$ check f_c and Δf.

3. Keyword Explanation

1. Frequency mixer (converter)

↩ back

A multiplier followed by a bandpass filter that shifts a signal's carrier to the sum or the difference of its frequency and the LO frequency.

4. Notes and Remarks

- Lathi (Sec. 5.3, pp. 225–226, Fig. 5.10) describes the same structure for a 91.2 MHz transmitter.
- The NBFM deviation of 5 Hz is kept small so that $\beta \ll 1$ even at the lowest audio frequency, which keeps the narrowband approximation (and its distortion) small.

5. Final Concise Answer

(a) $\varphi_{SSB} = m \cos \omega_c t \mp m_h \sin \omega_c t$: one branch DSB-SC, the other a $-90°$ shifted message times $\sin \omega_c t$; subtracting cancels the LSB (USB out), adding cancels the USB (LSB out).

(b) Message amplitude sets Δf: more significant lines and a nearly proportional rise in bandwidth. Message frequency sets the line spacing and lowers β: fewer lines, bandwidth rises only slightly.

(c) $\times 16$ (4 doublers) $\rightarrow$ mixer, $f_{LO} = 440.15625$ kHz, sum 760.15625 kHz $\rightarrow$ $\times 128$ (7 doublers): 97.3 MHz, 10.24 kHz.

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4. (a) In an FM system the frequency deviation is 5 kHz, when its audio modulating frequency is 400 Hz, and the audio modulating voltage is 2 V. Compute the modulation index. Also compute the frequency deviation and the modulation index if (i) if voltage is increased to 6V keeping modulation frequency unaltered, (ii) if voltage is increased to 8 V while the modulating frequency is reduced to 200 Hz. Marks: 5

(b) Explain slope detection method of FM demodulation. Marks: 6

(c) Define and explain uniform and non-uniform quantization. Marks: 4

(d) Five telemetry signals, each of bandwidth 2400 Hz, are to be transmitted simultaneously by binary PCM. The signals must be sampled at least 20% above the Nyquist rate. Framing and synchronizing requires an additional 0.5% extra bits. A PCM encoder is used to convert these signals before they are time-multiplexed into a single data stream. Determine the minimum possible data rate (bit per second) that must be transmitted, and the minimum bandwidth required to transmit the multiplexed signal. Marks: 5

(e) Show that the receiver of DM is essentially an accumulator. Marks: 5

4.(a) In an FM system the frequency deviation is 5 kHz, when its audio modulating frequency is 400 Hz, and the audio modulating voltage is 2 V. Compute the modulation index. Also compute the frequency deviation and the modulation index if (i) if voltage is increased to 6V keeping modulation frequency unaltered, (ii) if voltage is increased to 8 V while the modulating frequency is reduced to 200 Hz. Marks: 5

1. Theory

1. Deviation and modulation index (Core Idea)

In FM the carrier frequency swings up and down with the message: $f_i = f_c + k_f m(t)$. For a tone of peak voltage V_m and frequency f_m:

- the **frequency deviation** (largest swing) is $\Delta f = k_f V_m$: it is proportional to the **voltage** (amplitude) of the message and does not depend on its frequency;
- the **modulation index** is $\beta = \Delta f / f_m$: how many times the swing is larger than the message frequency.

The constant k_f (Hz per volt) belongs to the modulator, so it stays the same when the message changes.

2. Answer

Note on the paper

The paper prints "if voltage frequency is increased to 6V" with "frequency" struck through, so case (i) reads "if voltage is increased to 6 V".

Given: $\Delta f = 5$ kHz at $V_m = 2$ V and $f_m = 400$ Hz.

Step 1: Modulation index and sensitivity

$$\beta = \frac{\Delta f}{f_m} = \frac{5000}{400} = 12.5$$

$$k_f = \frac{\Delta f}{V_m} = \frac{5000}{2} = 2500 \text{ Hz/V}$$

Step 2: Case (i), 6 V at 400 Hz

$$\Delta f = k_f V_m = 2500 \times 6 = 15\,000 \text{ Hz} = 15 \text{ kHz}, \qquad \beta = \frac{15\,000}{400} = 37.5$$

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Step 3: Case (ii), 8 V at 200 Hz

$$\Delta f = 2500 \times 8 = 20\,000 \text{ Hz} = 20 \text{ kHz}, \qquad \beta = \frac{20\,000}{200} = 100$$

$$\boxed{\beta = 12.5; \quad \text{(i) } \Delta f = 15 \text{ kHz}, \ \beta = 37.5; \quad \text{(ii) } \Delta f = 20 \text{ kHz}, \ \beta = 100}$$

Final Answer: tripling the voltage triples the deviation and the index; in (ii) the voltage raises Δf fourfold and halving f_m doubles β again, to 100.

3. Notes and Remarks

- The modulating frequency affects only β, never Δf, in FM. In PM it would affect both.

4.(b) Explain slope detection method of FM demodulation. Marks: 6

1. Theory

1. What an FM demodulator must do (Core Idea)

An FM signal carries the message in its **instantaneous frequency**; its amplitude is constant. A demodulator must give an output voltage proportional to the frequency, $v_o \propto f_i - f_c = k_f m(t)$. The usual trick is to first turn the frequency changes into **amplitude** changes (FM to AM), then use a simple envelope detector.

2. Differentiation turns FM into AM

The derivative of a sinusoid multiplies its amplitude by its frequency: $\frac{d}{dt}\cos\omega t = -\omega \sin\omega t$. If the frequency changes with time, so does the amplitude of the derivative. Any circuit whose gain rises linearly with frequency over the band does the same job, and a tuned (resonant) circuit's sloping side is such a circuit.

2. Answer

Below, Step 1 is marked **Basic** (the derivative of an FM signal); a student who knows it can skip it.

Step 1 (Basic): Differentiating an FM signal

$$\varphi_{FM}(t) = A\cos\left[\omega_c t + k_f \int m(\alpha)\, d\alpha\right]$$

By the chain rule (the derivative of the angle is $\omega_c + k_f m(t)$):

$$\dot{\varphi}_{FM}(t) = -A\left[\omega_c + k_f m(t)\right] \sin\left[\omega_c t + k_f \int m(\alpha)\, d\alpha\right]$$

This is an AM-like signal whose envelope is $A[\omega_c + k_f m(t)]$; since $\omega_c > k_f|m|$ the envelope never goes negative.

Step 2: Envelope detection

An envelope detector gives $A[\omega_c + k_f m(t)]$; a capacitor that blocks the dc part $A\omega_c$ leaves

$$v_o(t) = Ak_f\, m(t) \ \propto \ m(t)$$

Step 3: The slope detector

An ideal differentiator is not needed: a **tuned circuit** whose resonant frequency f_0 is set a little **above** f_c works the same way over a small band. The carrier sits on the rising **slope** of its response $|H(f)|$, where the response is nearly a straight line.

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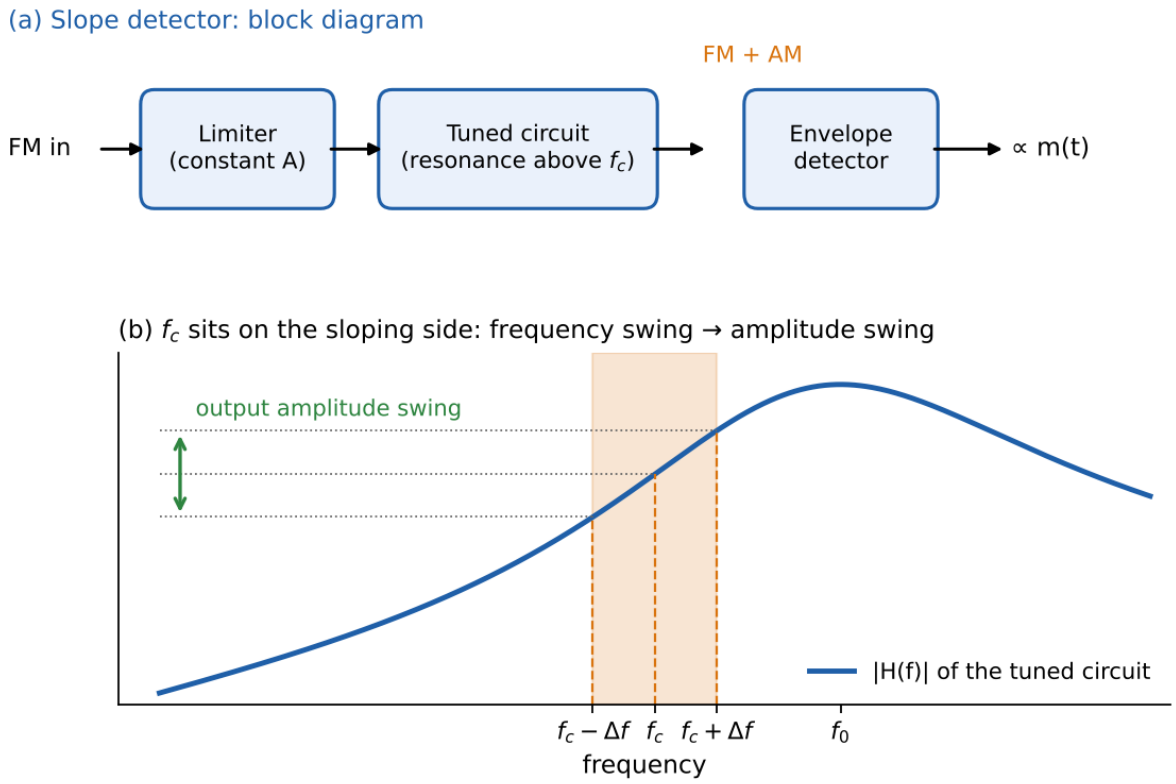


Figure 10: Slope detector: (a) blocks; (b) the carrier on the sloping side of the tuned circuit's response turns frequency swing into amplitude swing

Reading the figure. In (a) the FM signal passes a limiter, then the off-tuned circuit, whose output (orange label) is FM with amplitude variations, then an envelope detector whose output follows $m(t)$. In (b) the blue curve is the tuned circuit's response, peaking at f_0. The orange band marks the FM swing from $f_c - \Delta f$ to $f_c + \Delta f$ on the rising side; the dotted lines show that this frequency swing makes the output amplitude swing up and down (green arrow).

- When f_i rises towards f_0, the circuit's gain rises, so the output amplitude rises; when f_i falls, the amplitude falls. The amplitude therefore follows $m(t)$.
- The **limiter** in front removes amplitude variations caused by noise, which the slope circuit would otherwise pass to the output.

Step 4: Limitations and the balanced slope detector

- The slope of a single tuned circuit is linear only over a small range, so a large deviation gives **distortion**.
- The dc level and the response depend on the tuning.
- The **balanced slope detector** uses two tuned circuits, one above and one below f_c, with two envelope detectors whose outputs are subtracted. The curvature of the two slopes cancels, giving a wider linear range and no dc at f_c.

Steps to remember

FM → differentiate (or off-tuned slope) → envelope $A(\omega_c + k_f m)$ → envelope detector → block dc → $\propto m(t)$; limiter first, balanced version for linearity.

3. Keyword Explanation

1. Limiter

↩ back

A circuit that clips the signal to a constant amplitude, removing amplitude noise while keeping the zero crossings (the frequency information).

4. Notes and Remarks

- Lathi (Sec. 5.4) presents the ideal differentiator followed by an envelope detector and notes that any system with $|H(f)|$ linear over the FM band can replace the differentiator; the slope detector is that system.

4.(c) Define and explain uniform and non-uniform quantization. Marks: 4

1. Theory

1. Why quantize? (Core Idea)

A sampled signal can take any value. To send it as bits, each sample is rounded to the nearest of a fixed set of L levels: this is **quantization**. The rounding error is the **quantization noise**.

2. Answer

Quantization is the rounding of each sample of a signal of range $-m_p$ to $+m_p$ to the nearest of L allowed levels.

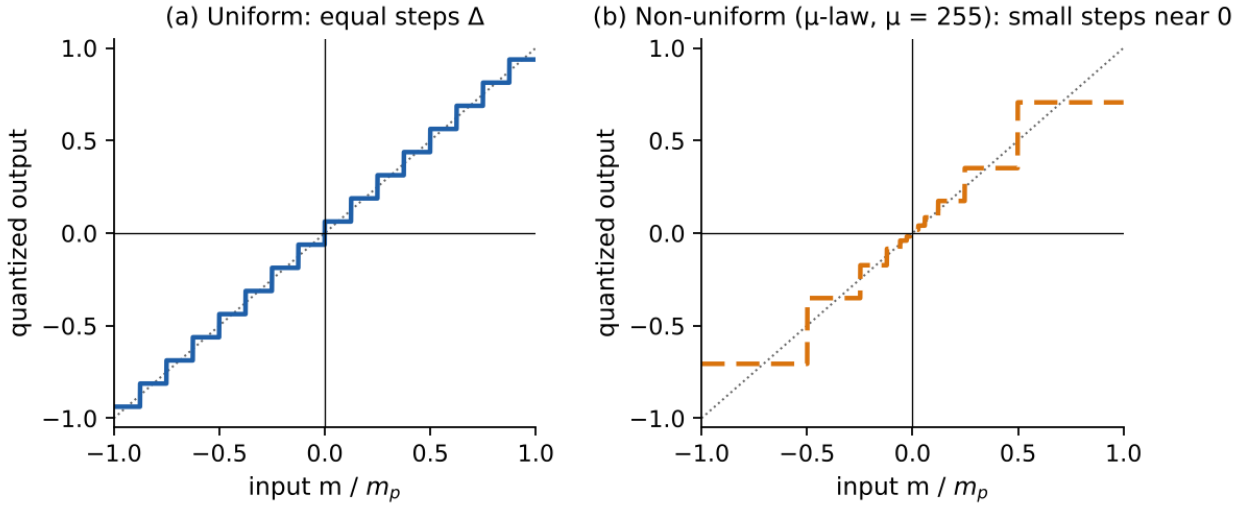


Figure 11: Quantizer characteristics with 16 levels: (a) uniform, equal steps; (b) non-uniform (μ-law), small steps near zero and large steps near the peak

Reading the figure. Both plots show output against input, with the grey dotted line marking a perfect copy. In (a) the solid blue staircase has steps of equal size everywhere. In (b) the dashed orange staircase has tiny steps near zero and large steps near $\pm m_p$.

Uniform quantization

- The levels are **equally spaced**, with step $\Delta = 2m_p/L$.
- The error is at most $\pm\Delta/2$ for every sample, and the noise power is $\Delta^2/12 = m_p^2/(3L^2)$, the same for weak and strong signals.
- So the **SQNR** $= 3L^2\,\overline{m^2}/m_p^2$ is good for loud signals but poor for quiet ones (speech is mostly quiet).

Non-uniform quantization

- The steps are **small near zero** and **large near the peak**, so the error is roughly a fixed fraction of the signal's own size, and the SQNR is nearly the same for weak and strong signals.
- In practice it is done by **companding**: a compressor (μ-law, $y = \dfrac{\ln(1+\mu|m|/m_p)}{\ln(1+\mu)}\mathrm{sgn}(m)$, $\mu = 255$ in North America; or A-law, $A = 87.6$ in Europe) boosts small amplitudes, a uniform quantizer follows, and an expander at the receiver undoes the compression.

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Final Summary: uniform = equal steps, constant noise, SQNR falls for weak signals; non-uniform = steps growing with amplitude, nearly constant SQNR, made by companding (μ-law, A-law).

3. Notes and Remarks

- Lathi, Sec. 6.2.

4.(d) Five telemetry signals, each of bandwidth 2400 Hz, are to be transmitted simultaneously by binary PCM. The signals must be sampled at least 20% above the Nyquist rate. Framing and synchronizing requires an additional 0.5% extra bits. A PCM encoder is used to convert these signals before they are time-multiplexed into a single data stream. Determine the minimum possible data rate (bit per second) that must be transmitted, and the minimum bandwidth required to transmit the multiplexed signal. Marks: 5

1. Theory

1. The PCM chain (Core Idea)

Each signal is **sampled**, each sample **quantized** to $L = 2^n$ levels and **coded** in n bits; the bits of all signals are then **time-multiplexed** into one stream.

- **Nyquist rate:** a signal of bandwidth B must be sampled at least $2B$ times per second.
- **Bit rate:** (number of signals) $\times$ (samples per second) $\times$ (bits per sample), plus any framing bits.
- **Minimum bandwidth:** a binary stream of R bits/s needs at least $R/2$ Hz (Nyquist's minimum bandwidth for pulses sent without interference).

2. Answer

Note on the paper

The paper does not say how many bits each sample gets (Lathi's Problem 6.2-4, of the same wording, does not either). **Assumption:** $n = 8$ bits per sample (256 levels), the usual telephony value; the general result for n bits is given too.

Given: 5 signals, $B = 2400$ Hz each, sampling 20 % above Nyquist, 0.5 % extra bits, $n = 8$.

Step 1: Sampling rate

$$f_{Nyq} = 2B = 4800 \text{ Hz}, \qquad f_s = 1.2 \times 4800 = 5760 \text{ samples/s per signal}$$

Step 2: Bit rate before framing

$$r = 5 \times 5760 \times n = 28\,800\,n = 28\,800 \times 8 = 230\,400 \text{ bits/s}$$

Step 3: Add 0.5 % framing and synchronizing bits

$$R = 1.005 \times 230\,400 = 231\,552 \text{ bits/s}$$

$$\boxed{R \approx 231.6 \text{ kbps}}$$

Step 4: Minimum bandwidth

$$B_T = \frac{R}{2} = \frac{231\,552}{2} = 115\,776 \text{ Hz}$$

$$\boxed{B_T \approx 115.8 \text{ kHz}}$$

General result: $R = 28\,944\,n$ bits/s and $B_T = 14\,472\,n$ Hz; for example with $n = 9$ bits, 260.5 kbps and 130.2 kHz.
Final Answer: with 8 bits per sample, $R \approx 231.6$ kbps and $B_T \approx 115.8$ kHz.

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3. Notes and Remarks

- If a maximum quantization error were given (say e of the peak), n would follow from $m_p/L \le e\,m_p$, so $L \ge 1/e$; this is how the ECG problem of the 2020-21 and 2021-22 papers sets n.

4.(e) Show that the receiver of DM is essentially an accumulator. Marks: 5

1. Theory

1. Delta modulation (Core Idea)

Delta modulation (DM) sends **one bit per sample**: only whether the signal is above or below the transmitter's own running estimate of it. The estimate then moves one fixed step σ up or down. Because the samples are taken very often (many times the Nyquist rate), the signal changes little between samples, and one bit is enough to follow it.

2. Answer

Below, the equations follow Lathi (Sec. 6.7).

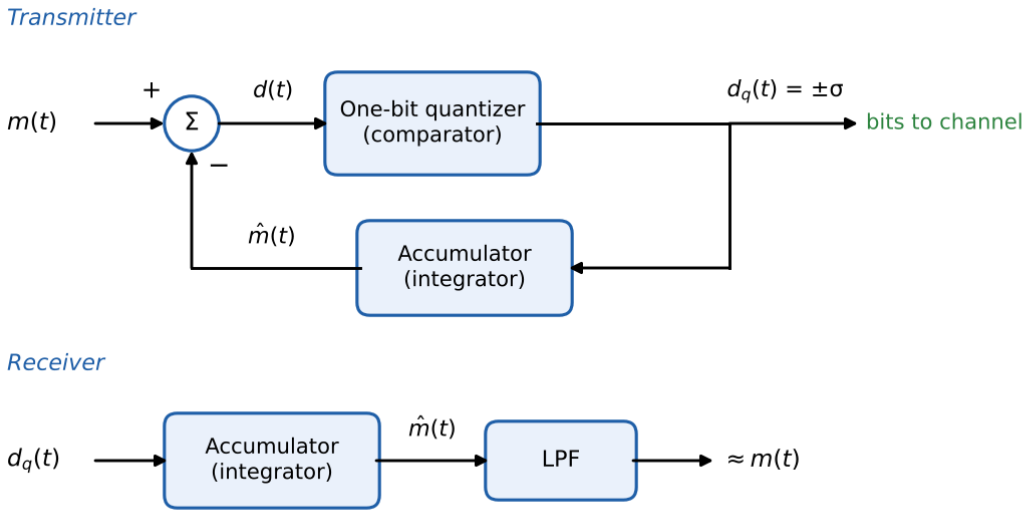


Figure 12: DM transmitter and receiver: the transmitter's feedback and the receiver both use an accumulator (reused from the 2020-21 paper)

Reading the figure. At the transmitter $m(t)$ is compared with the estimate $\hat{m}(t)$; the one-bit quantizer sends $d_q = \pm\sigma$ to the channel, and the same d_q goes into an accumulator to make the next estimate. The receiver is just the same accumulator followed by an LPF.

Step 1: The transmitter's equations

Let $m[k]$ be the k-th sample and $\hat{m}_q[k]$ the estimate after it. The difference and its one-bit quantized value are

$$d[k] = m[k] - \hat{m}_q[k-1], \qquad d_q[k] = \sigma\,\mathrm{sgn}(d[k]) = \pm\sigma$$

and the estimate is updated by the transmitted step:

$$\hat{m}_q[k] = \hat{m}_q[k-1] + d_q[k]$$

Step 2: Unroll the update

Apply the update again to $\hat{m}_q[k-1]$, then to $\hat{m}_q[k-2]$, and so on:

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$$\hat{m}_q[k] = \hat{m}_q[k-2] + d_q[k-1] + d_q[k] = \cdots = \hat{m}_q[-1] + \sum_{i=0}^{k} d_q[i]$$

Step 3: Start from zero

With the estimate starting at $\hat{m}_q[-1] = 0$:

$$\boxed{\hat{m}_q[k] = \sum_{i=0}^{k} d_q[i]}$$

The estimate is simply the **running sum** of all the transmitted steps.

Step 4: The receiver

The receiver gets exactly the $d_q[k]$ (if there are no channel errors). To rebuild $\hat{m}_q[k]$ it needs only to add them up: that is an **accumulator** (a summer with a one-sample delay in feedback, or, for a train of $\pm\sigma$ pulses, an integrator). An LPF then smooths the staircase to give $m(t)$. So the DM receiver is essentially an accumulator.

Steps to remember

$d_q[k] = \pm\sigma \to \hat{m}_q[k] = \hat{m}_q[k-1] + d_q[k] \to$ unroll $\to \hat{m}_q[k] = \sum d_q[i] \to$ receiver = accumulator + LPF.

3. Keyword Explanation

1. Accumulator

↩ back

A circuit whose output is the running sum of its inputs: $y[k] = y[k-1] + x[k]$.

4. Notes and Remarks

- Because the receiver sums the bits, a channel error is never forgotten: it shifts all later values. Sigma-delta modulation removes this weakness.

5. Final Concise Answer

(a) $\beta = 12.5$; (i) $\Delta f = 15$ kHz, $\beta = 37.5$; (ii) $\Delta f = 20$ kHz, $\beta = 100$.

(b) A limiter, a tuned circuit tuned above f_c (its slope turns frequency into amplitude, like a differentiator), and an envelope detector; the balanced version uses two tuned circuits for linearity.

(c) Uniform: equal steps $2m_p/L$, constant noise, poor SQNR for weak signals; non-uniform: small steps near zero, near-constant SQNR, by μ-law or A-law companding.

(d) With 8 bits per sample: $R = 231.6$ kbps, $B_T = 115.8$ kHz (in general $28\,944n$ bit/s and $14\,472n$ Hz).

(e) $\hat{m}_q[k] = \hat{m}_q[k-1] + d_q[k] \Rightarrow \hat{m}_q[k] = \sum_{i \le k} d_q[i]$: the receiver just accumulates the received steps.

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5. (a) Explain principle of sigma-delta modulator. Also explain how it is advantageous over DM. Marks: 5
   (b) Briefly explain the desired characteristics of a line coding scheme. Draw the line coded signal for the information bits '1010011100' for the following line coding scheme- i) Manchester, ii) AMI, iii) Di-code. Marks: 7
   (c) Explain DPSK principle with necessary block diagram. Marks: 8
   (d) Describe coherent detection of BFSK. Marks: 5

5.(a) Explain principle of sigma-delta modulator. Also explain how it is advantageous over DM. Marks: 5

1. Theory

1. The weak points of DM (Core Idea)

DM follows the signal one step σ per sample. Two things go wrong:

- **Slope overload:** if the signal rises faster than σf_s, the staircase cannot keep up. The highest slope of a tone $A\cos\omega_m t$ is $A\omega_m$, so high-frequency tones overload most easily.
- **Error build-up:** the receiver adds up every received step, so one channel error shifts all later values.

2. The idea of sigma-delta

Integrate the message *before* delta modulation. The integral of a high-frequency tone is small (A/ω_m), so its slope is just A, whatever the frequency. At the receiver we would then need a differentiator after the DM accumulator; but differentiating an integral gives back the original, so the two cancel.

2. Answer

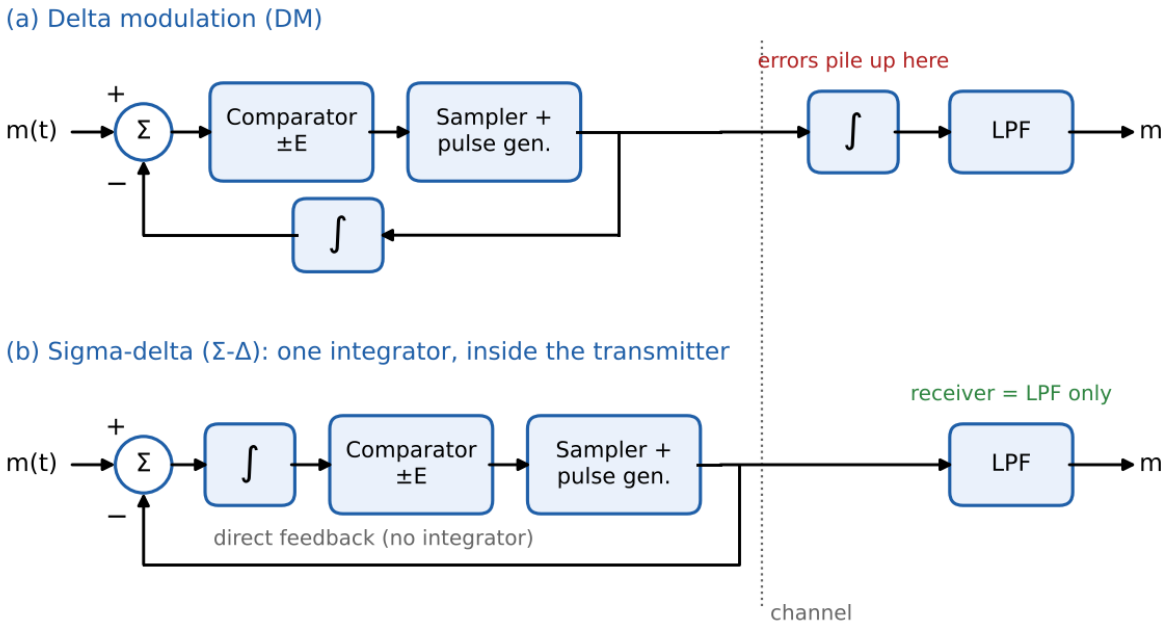


Figure 13: (a) DM; (b) sigma-delta: the integrator moved to the transmitter's input, leaving only an LPF at the receiver (reused from the 2021-22 paper)

Reading the figure. In (a), DM: the comparator output goes to the channel, the feedback has an integrator, and the receiver is an integrator followed by an LPF, where errors pile up. In (b), sigma-delta: one integrator sits after the difference node inside the transmitter, the feedback is direct, and the receiver is only an LPF.

Principle.

1. The message $m(t)$ is integrated and then delta-modulated. The integrator at the input (the Σ, "sum") and the DM integrator in the feedback are combined into a single integrator placed after the difference node (the Δ): $m(t)$ minus the feedback pulses is integrated, and a comparator decides $\pm E$ each sampling instant.
2. The output pulse train has a **density** of positive pulses that follows the **amplitude** of $m(t)$ (not its slope, as in DM).
3. At the receiver the DM integrator and the needed differentiator cancel, so the receiver is just a **low-pass filter** that averages the pulse density into $m(t)$.

Advantages over DM:

- **No frequency-dependent slope overload:** the overload depends on the amplitude only, so high-frequency message components are handled as well as low ones.
- **Simpler receiver:** only an LPF; no integrator.
- **No error build-up:** a channel error affects only the local average, not all later values.
- **Passes dc and low frequencies well,** and the quantization noise is pushed to high frequencies, where the LPF removes it (noise shaping), giving a better SNR in the message band. This is why sigma-delta is used in audio converters.

3. Notes and Remarks

- Lathi (Sec. 6.7) calls it delta-sigma modulation.

5.(b) Briefly explain the desired characteristics of a line coding scheme. Draw the line coded signal for the information bits '1010011100' for the following line coding scheme- i) Manchester, ii) AMI, iii) Di-code. Marks: 7

1. Theory

1. What is line coding? (Core Idea)

Bits are just 1s and 0s. To send them on a wire they must become voltage pulses. A **line code** is the rule that turns each bit (or each change of bit) into a voltage shape. Different rules give signals with different spectra, different help for the receiver's clock, and different abilities to catch errors.

2. The three codes asked

- **Manchester (split-phase):** every bit has a transition in the middle. Following Lathi and the course notes, a 1 is $+V$ for the first half and $-V$ for the second (high to low), and a 0 is the reverse (low to high).
- **AMI (alternate mark inversion, bipolar):** a 0 is 0 V; the 1s ("marks") are sent alternately as $+V$ and $-V$.
- **Dicode:** marks the **changes** in the data: a change from 0 to 1 is $-V$, a change from 1 to 0 is $+V$, and no change is 0 V (the course notes' rule).

2. Answer

Desired characteristics of a line code

1. **Small transmission bandwidth:** as narrow a spectrum as possible.
2. **No dc (and little low-frequency) power:** so the signal can pass transformers and ac-coupled repeaters, and the baseline does not wander.
3. **Enough timing content:** frequent transitions so the receiver can recover the clock, even for long runs of 0s or 1s (self-synchronization).
4. **Power efficiency:** low error probability for a given transmitted power.

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5. **Error detection** (and if possible correction) built into the code.
6. **Transparency:** correct operation for any bit pattern.

The waveforms for 1 0 1 0 0 1 1 1 0 0

Bit	1	0	1	0	0	1	1	1	0	0
Manchester	$+/-$	$-/+$	$+/-$	$-/+$	$-/+$	$+/-$	$+/-$	$+/-$	$-/+$	$-/+$
AMI	$+V$	0	$-V$	0	0	$+V$	$-V$	$+V$	0	0
Dicode	$-V$	$+V$	$-V$	$+V$	0	$-V$	0	0	$+V$	0

Table 3: Summary

Assumptions: the first 1 of AMI is positive; for dicode the bit before the sequence is taken as 0, so the first bit ($0 \to 1$) gives $-V$.

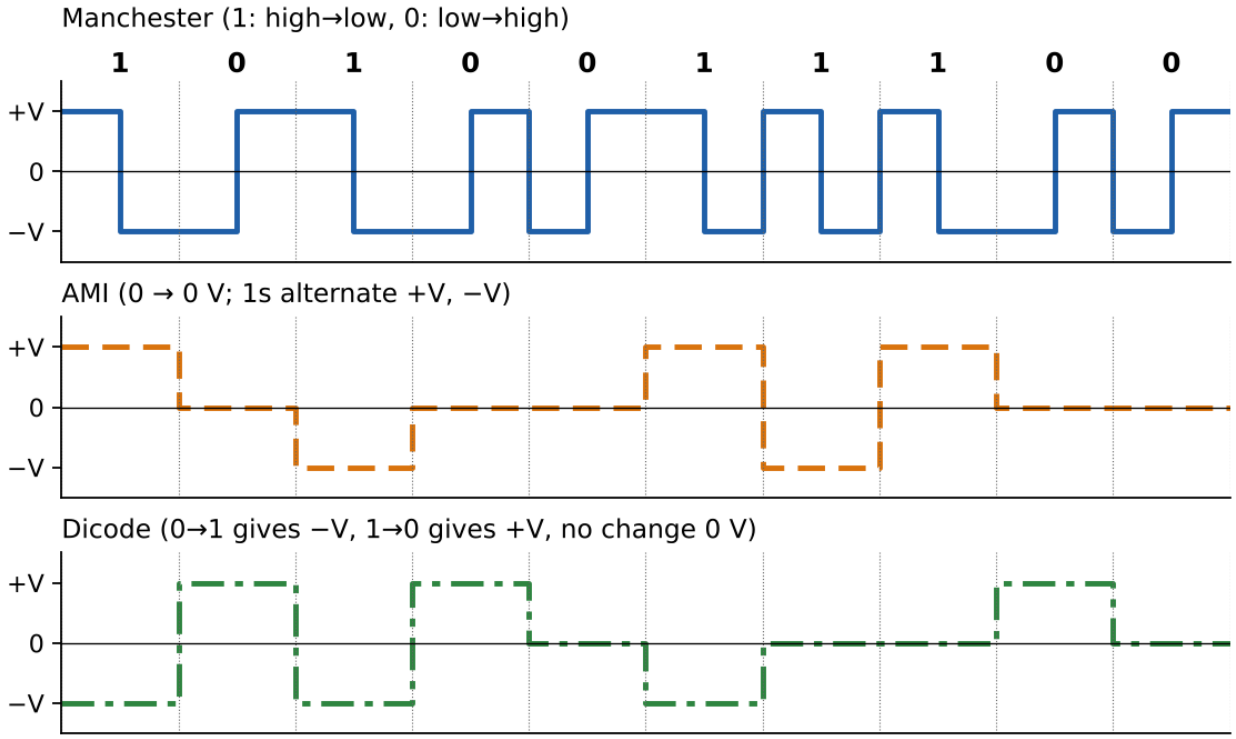


Figure 14: Manchester (solid blue), AMI (dashed orange) and dicode (dash-dot green) for 1010011100

Reading the figure. The bits are written above the top panel, and dotted lines mark the bit boundaries. Manchester (solid blue) changes level in the middle of every bit: high to low for 1, low to high for 0. AMI (dashed orange) is 0 V for each 0 and alternates $+V$, $-V$, $+V$, … for the 1s (bits 1, 3, 6, 7, 8). Dicode (dash-dot green) is non-zero only where the bit changes: $-V$ for $0 \to 1$ (bits 1, 3, 6) and $+V$ for $1 \to 0$ (bits 2, 4, 9).

Comparison: Manchester has no dc and a transition in every bit (self-clocking) but needs twice the bandwidth; AMI has no dc, low bandwidth and detects single errors (two same-polarity marks in a row), but loses timing in long runs of 0s; dicode has no dc and a pulse at every change, but nothing during long runs of equal bits.

3. Notes and Remarks

- IEEE 802.3 (Ethernet) uses the opposite Manchester convention (0 high to low, 1 low to high); state the convention in the exam.
- Lathi draws AMI with half-width (return-to-zero) pulses; the full-width (NRZ) form drawn here is Forouzan's and the course notes'. The pulse polarities are the same.

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- Some books (e.g. Sklar) give dicode the opposite polarity ($0 \to 1$ gives $+V$); the course notes' rule is used here.

5.(c) Explain DPSK principle with necessary block diagram. Marks: 8

1. Theory

1. Why DPSK? (Core Idea)

In **PSK** a 1 is sent as $+A\cos\omega_c t$ and a 0 as $-A\cos\omega_c t$. To tell them apart the receiver needs a local carrier with the exact phase of the transmitter's (coherent detection), which is difficult. **Differential PSK** puts the information in the **change** of phase from one bit to the next. The receiver then compares each received bit with the previous one, which it already has, so no local carrier is needed.

2. Differential encoding

The data bits I_k are first turned into encoded bits q_k by

$$q_k = I_k \oplus q_{k-1}$$

(XOR, as in Lathi): a data 1 flips the encoded bit, a data 0 keeps it. The encoded bits are then sent by ordinary PSK. So a data 1 means "phase changed by π", a data 0 means "phase unchanged".

2. Answer

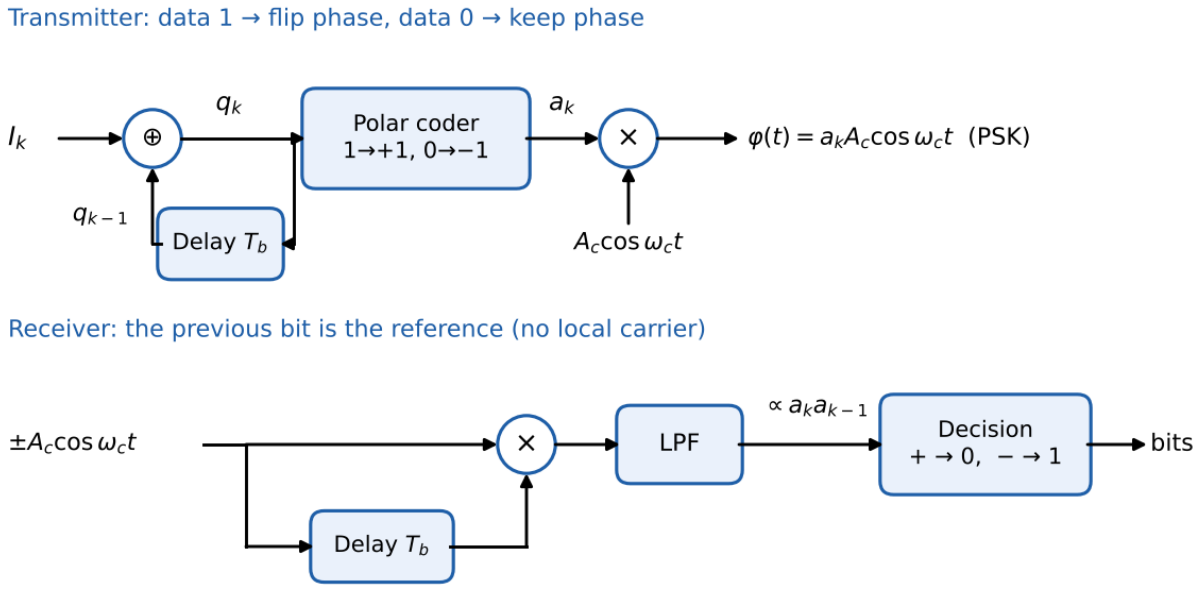


Figure 15: DPSK transmitter (XOR with the delayed encoded bit, then PSK) and receiver (multiply by the signal delayed one bit, LPF, decision); reused from the 2021-22 paper

Reading the figure. In the transmitter, the data bit I_k and the previous encoded bit q_{k-1} (from the delay T_b) go into the XOR; the encoded bit q_k is made polar ($a_k = \pm 1$) and multiplies the carrier. In the receiver, the incoming signal is multiplied by itself delayed by one bit, low-pass filtered to give a value proportional to $a_k a_{k-1}$, and the decision gives 0 for a positive value and 1 for a negative one.

Step 1: Encoder

$q_k = I_k \oplus q_{k-1}$, starting from a reference bit q_0; then $a_k = +1$ for $q_k = 1$ and -1 for $q_k = 0$.

Step 2: Modulator

$\varphi(t) = a_k A\cos\omega_c t$ during bit k: plain PSK of the encoded bits.

Step 3: Receiver

Multiply the received bit by the previous received bit (a delay of T_b):

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$$a_k A\cos\omega_c t \times a_{k-1} A\cos\omega_c t = \frac{a_k a_{k-1} A^2}{2}(1+\cos 2\omega_c t)$$

The LPF removes the $2\omega_c$ term, leaving $\frac{1}{2}a_k a_{k-1}A^2$:

- positive if $a_k = a_{k-1}$ (no phase change): data **0**,
- negative if $a_k \neq a_{k-1}$ (phase change): data **1**.

Step 4: Example

Data 1011001 with reference $q_0 = 0$:

k	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Data I_k		1	0	1	1	0	0	1
Encoded q_k	0 (ref)	1	1	0	1	1	1	0
Phase	π	0	0	π	0	0	0	π
Phase changed?		yes	no	yes	yes	no	no	yes
Decoded		1	0	1	1	0	0	1

Table 4: Summary

The decoded bits equal the data.

Advantages: no carrier recovery (simple receiver); a constant phase offset of the channel cancels in the comparison. **Disadvantages:** the reference is the noisy previous bit, so it needs about 1 dB more E_b/N_0 than coherent PSK for the same error rate, and an error in one bit tends to cause errors in pairs.

Steps to remember

$q_k = I_k \oplus q_{k-1} \rightarrow$ PSK of $q_k \rightarrow$ receiver multiplies by the 1-bit-delayed signal $\rightarrow$ LPF $\propto a_k a_{k-1} \rightarrow$ $+$ means 0, $-$ means 1.

3. Notes and Remarks

- Lathi, Sec. 7.8 (differential PSK).

5.(d) Describe coherent detection of BFSK. Marks: 5

1. Theory

1. What is BFSK? (Core Idea)

Binary frequency-shift keying sends a 1 as a burst of one frequency and a 0 as a burst of another:

$$s(t) = \begin{cases} A\cos\omega_1 t, & \text{bit } 1 \\ A\cos\omega_2 t, & \text{bit } 0 \end{cases} \qquad 0 \le t \le T_b$$

2. Correlation

To find out which frequency is present, multiply the signal by each candidate and integrate over one bit. A sinusoid times itself has a positive average; times a sinusoid of a well-chosen different frequency it averages to zero (the two are **orthogonal**).

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2. Answer

Coherent detection uses two local carriers, $\cos\omega_1 t$ and $\cos\omega_2 t$, locked in frequency and phase to the transmitter's, and two correlators.

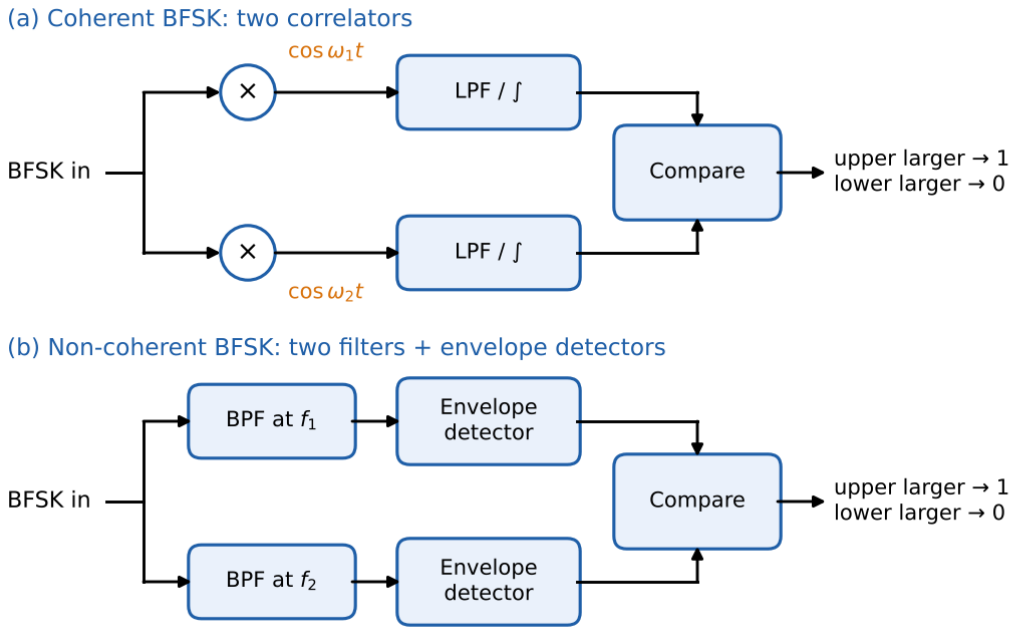


Figure 16: Coherent BFSK detector (a): two correlators and a comparator; (b) is the non-coherent version for comparison (reused from the 2021-22 paper)

Reading the figure. In (a) the BFSK input splits into two branches. The upper branch multiplies by $\cos\omega_1 t$, the lower by $\cos\omega_2 t$ (orange), each followed by an LPF or integrator over one bit; the comparator decides 1 if the upper output is larger and 0 if the lower is larger. Part (b) replaces the multipliers by bandpass filters and envelope detectors (non-coherent).

Step 1: Bit 1 received, upper branch

$$\int_0^{T_b} A\cos\omega_1 t\cdot\cos\omega_1 t\,dt = \frac{A}{2}\int_0^{T_b}(1+\cos 2\omega_1 t)\,dt \approx \frac{AT_b}{2}$$

(the $\cos 2\omega_1 t$ term integrates to nearly zero over many cycles).

Step 2: Bit 1 received, lower branch

$$\int_0^{T_b} A\cos\omega_1 t\cdot\cos\omega_2 t\,dt = \frac{A}{2}\int_0^{T_b}[\cos(\omega_1-\omega_2)t+\cos(\omega_1+\omega_2)t]\,dt \approx 0$$

if the two frequencies are orthogonal, that is $f_1 - f_2$ is a multiple of $1/(2T_b)$ (with the sum term again negligible).

Step 3: Decision

For bit 1 the upper output ($AT_b/2$) is larger; for bit 0 the lower one is. The comparator (in effect, the sign of upper minus lower) gives the bit.

Key Points:

- Coherent BFSK needs two phase-locked local carriers, so it is more complex than the non-coherent detector of (b), but needs about 1 dB less signal power for the same error rate.
- Its error probability is $P_b = Q\left(\sqrt{E_b/N_0}\right)$, 3 dB worse than coherent PSK, because the two signals are orthogonal rather than opposite.

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3. Notes and Remarks

- Lathi, Sec. 7.8.

4. Final Concise Answer

(a) Integrate the message, then delta-modulate (one integrator after the difference node); the pulse density follows the amplitude; the receiver is an LPF. Advantages: no frequency-dependent slope overload, simple receiver, no error build-up, noise shaping.

(b) Desired: small bandwidth, no dc, timing content, power efficiency, error detection, transparency. Manchester $+/-$ for 1, $-/+$ for 0; AMI $+, 0, -, 0, 0, +, -, +, 0, 0$; dicode $-, +, -, +, 0, -, 0, 0, +, 0$ (V).

(c) $q_k = I_k \oplus q_{k-1}$, PSK of q_k; the receiver multiplies by the signal delayed by T_b and low-pass filters: positive $\to$ 0, negative $\to$ 1.

(d) Two correlators with $\cos \omega_1 t$ and $\cos \omega_2 t$; the matching branch gives $AT_b/2$, the other about 0; the larger decides the bit.

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6. (a) With proper block diagram, describe modulation of QPSK. Also explain how it offers advantages over BPSK? Marks: 8
   (b) A multiplexer combines four 100-kbps channels using a time slot of 2 bits. Show the output for the following inputs- '110010', '001010', '101101', '000111'. What is the frame rate? What is the frame duration? What is the bit rate? What is the bit duration? Marks: 6
   (c) Briefly explain frequency division multiplexing techniques. Marks: 5
   (d) Write short notes on following terms-
      (i) Pulse stuffing
      (ii) Multi-slot multiplexing Marks: 6

6.(a) With proper block diagram, describe modulation of QPSK. Also explain how it offers advantages over BPSK? Marks: 8

1. Theory

1. From BPSK to QPSK (Core Idea)

BPSK sends one bit per symbol as $\pm A\cos\omega_c t$. A carrier also has a second, independent "direction": $\sin\omega_c t$ is orthogonal to $\cos\omega_c t$ (their product averages to zero). **QPSK** uses both at once: one bit on the cosine (**in-phase**, I) and one bit on the sine (**quadrature**, Q). So each symbol carries *two* bits, and the receiver can still separate them.

2. Answer

QPSK (quadrature phase-shift keying) sends two bits per symbol by choosing one of four carrier phases.

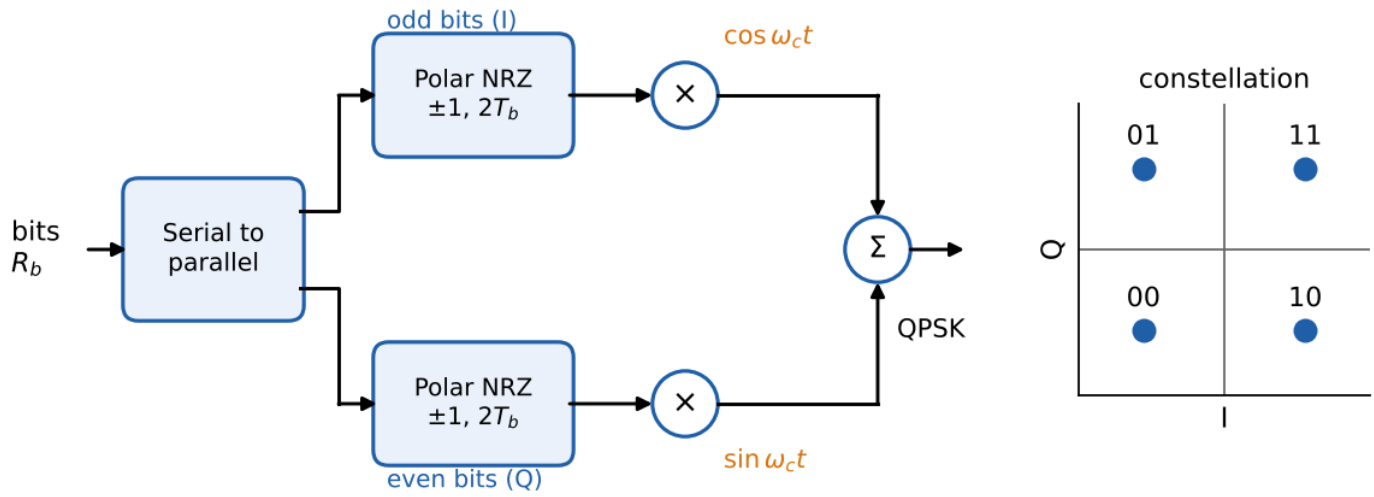


Figure 17: QPSK modulator and its constellation

Reading the figure. The serial bits (rate R_b) are split into two streams: odd bits to the I branch and even bits to the Q branch, each now at half the rate. Each stream is made polar (± 1) with pulses of length $2T_b$, multiplied by $\cos\omega_c t$ (I) or $\sin\omega_c t$ (Q), and the two products are added. The constellation on the right shows the four possible points, one per bit pair.

Step 1: Serial to parallel

The bit stream is split into pairs; the first bit of each pair drives the I branch, the second the Q branch. Each branch runs at $R_b/2$, with symbol length $T_s = 2T_b$.

Step 2: Polar coding

Each branch bit becomes $a_I, a_Q = +1$ (bit 1) or -1 (bit 0).

Step 3: Quadrature modulation and sum

$$\varphi_{QPSK}(t) = a_I A\cos\omega_c t + a_Q A\sin\omega_c t = \sqrt{2}A\cos(\omega_c t - \theta), \qquad \theta = \tan^{-1}\frac{a_Q}{a_I}$$

so the four bit pairs give four phases 90° apart (45°, 135°, 225°, 315°), all with the same amplitude $\sqrt{2}A$.

Advantages over BPSK:

- **Half the bandwidth for the same bit rate:** the symbol rate is $R_b/2$, so the main-lobe bandwidth is R_b instead of $2R_b$ (minimum $R_b/2$ instead of R_b). Equivalently, twice the bit rate in the same bandwidth.
- **Same error performance per bit:** the I and Q branches are two independent BPSK systems on orthogonal carriers, so the bit error probability for a given E_b/N_0 is the same as BPSK, $Q(\sqrt{2E_b/N_0})$.
- **Constant envelope:** like BPSK, the amplitude stays fixed, so efficient nonlinear power amplifiers can be used.

The cost is a more complex modulator and receiver (two branches and carrier recovery at 90°).

3. Notes and Remarks

- Lathi, Sec. 7.8 and 7.9 (M-ary PSK, $M = 4$).

6.(b) A multiplexer combines four 100-kbps channels using a time slot of 2 bits. Show the output for the following inputs- '110010', '001010', '101101', '000111'. What is the frame rate? What is the frame duration? What is the bit rate? What is the bit duration? Marks: 6

1. Theory

1. Synchronous TDM (Core Idea)

In **time-division multiplexing** several channels share one link by taking turns. Each turn is a **time slot**; one round of slots, one for each channel, is a **frame**. Here every slot carries 2 bits.

- **Frame rate** = how many frames per second = (bits per second of one channel) / (bits per slot).
- **Frame duration** = 1 / frame rate.
- **Link bit rate** = frame rate × bits per frame.
- **Bit duration** = 1 / link bit rate.

2. Answer

Given: 4 channels, 100 kbps each, 2 bits per slot.
Assumption: each input is read from left to right, so its leftmost bits are sent first.

Step 1: The output

Each frame takes 2 bits from channel 1, then 2 from channel 2, 3 and 4:

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Frame	Ch 1	Ch 2	Ch 3	Ch 4	Frame bits
1	11	00	10	00	11001000
2	00	10	11	01	00101101
3	10	10	01	11	10100111

Table 5: Summary

Output stream: **11001000 00101101 10100111**.

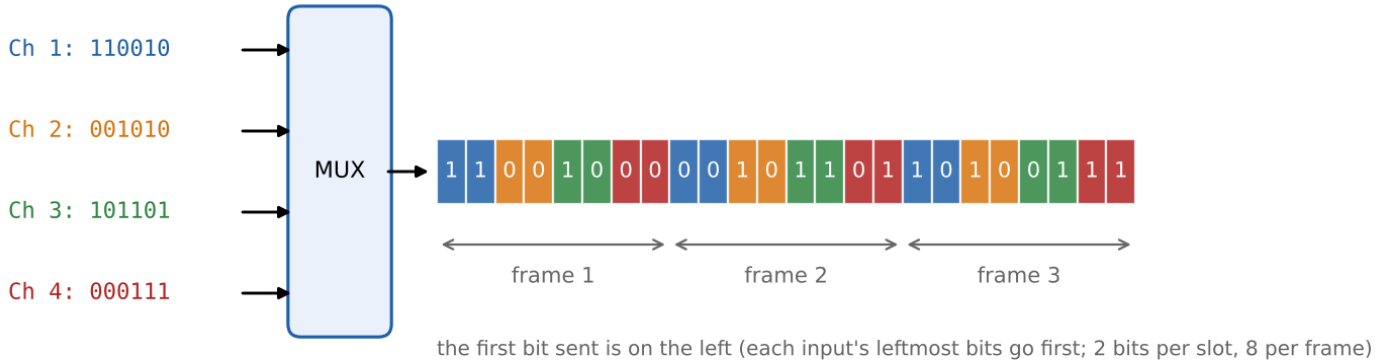


Figure 18: TDM output: three frames of four 2-bit slots, coloured by channel

Reading the figure. The four inputs on the left (blue, orange, green, red) enter the MUX; the output on the right is a row of bits coloured by the channel they came from, two bits per slot, four slots (8 bits) per frame, three frames in all.

Step 2: Frame rate

Each frame takes 2 bits of each channel, and each channel delivers 100 000 bits/s:

$$\text{frame rate} = \frac{100\,000}{2} = 50\,000 \text{ frames/s}$$

Step 3: Frame duration

$$T_f = \frac{1}{50\,000} = 20\ \mu\text{s}$$

Step 4: Bit rate

$$R = 50\,000 \times (4 \times 2) = 400\,000 \text{ bits/s} = 400 \text{ kbps}$$

Step 5: Bit duration

$$T_b = \frac{1}{400\,000} = 2.5\ \mu\text{s}$$

$$\boxed{50\,000 \text{ frames/s}, \quad 20\ \mu\text{s}, \quad 400 \text{ kbps}, \quad 2.5\ \mu\text{s}}$$

Final Answer: output 11001000 | 00101101 | 10100111; frame rate 50 000 frames/s; frame duration 20 μs; bit rate 400 kbps; bit duration 2.5 μs.

3. Notes and Remarks

- Forouzan's figures draw the streams flowing right to left, so the rightmost bit enters first; read that way the frames are taken from the right-hand ends (frame 1 = 10 10 01 11, …). The rates do not change.
- Check: the link rate equals the sum of the inputs, $4 \times 100 = 400$ kbps, as it must with no extra bits.

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6.(c) Briefly explain frequency division multiplexing techniques. Marks: 5

1. Theory

1. Sharing a link by frequency (Core Idea)

A link usually has much more bandwidth than one signal needs. **FDM** lets several signals use it at the same time by giving each one its own band of frequencies, like radio stations sharing the air.

2. Answer

FDM is an analog multiplexing technique in which each signal modulates its own carrier frequency, and the modulated signals, placed side by side in frequency, are combined into one composite signal on the link.

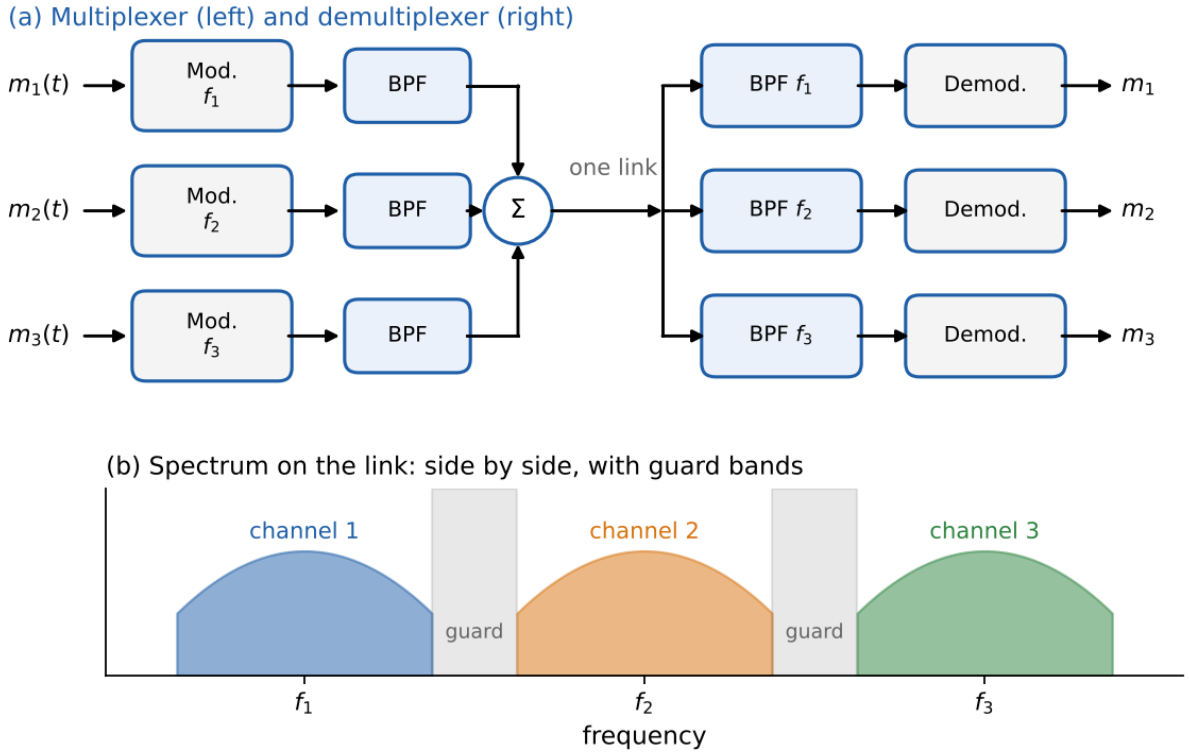


Figure 19: FDM: (a) multiplexer and demultiplexer; (b) the channels side by side on the link with guard bands

Reading the figure. In (a) each message m_1, m_2, m_3 modulates its own carrier f_1, f_2, f_3; band-limiting filters keep each in its slot, and the summer puts them on one link. At the receiver bandpass filters at f_1, f_2, f_3 separate the channels and demodulators recover the messages. In (b) the three channels sit side by side, separated by grey guard bands.

- **Multiplexing:** each source's signal is modulated (AM, SSB or FM) onto a different carrier; the carriers are spaced so the channels do not overlap.
- **Guard bands:** unused strips of frequency between channels prevent overlap and allow practical filters.
- **Demultiplexing:** bandpass filters separate the channels, and each is demodulated.
- **Analog hierarchy (telephone):** 12 voice channels make a **group** (48 kHz); 5 groups a **supergroup** (240 kHz); 10 supergroups a **master group** (2.52 MHz); 6 master groups a **jumbo group** (16.984 MHz).
- **Applications:** AM and FM radio broadcasting, television, first-generation cellular telephones; WDM in optical fibre is the same idea with light.
- **Total bandwidth:** $B_{total} = N \times B_{channel} + (N-1) \times B_{guard}$.

Pros and cons: simple and continuous (no synchronization of slots), but nonlinearity in the link causes crosstalk between channels, and guard bands waste bandwidth.

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3. Notes and Remarks

- Forouzan, Sec. 6.1.

6.(d) Write short notes on following terms- (i) Pulse stuffing (ii) Multi-slot multiplexing Marks: 6

1. Theory

1. TDM with unequal rates (Core Idea)

Simple synchronous TDM gives every channel one slot of the same size per frame, so it works only if all inputs have the same data rate. Forouzan describes three ways to handle **unequal** rates: multilevel multiplexing, **multiple-slot allocation** and **pulse stuffing**.

2. Answer

(i) Pulse stuffing

When the input rates are **not integer multiples** of each other, the slower inputs are padded up to the rate of the fastest. The multiplexer inserts **dummy bits** (stuffing bits) into the slower streams so that all have the same rate, then multiplexes them normally. The receiver knows where the dummy bits are and discards them.

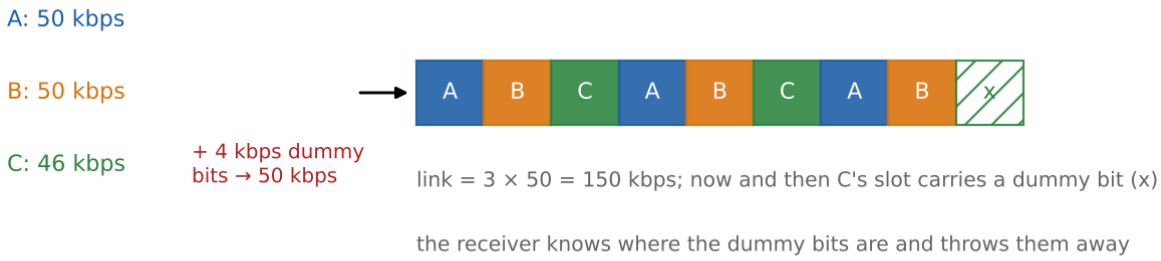


Figure 20: Pulse stuffing: channel C (46 kbps) gets 4 kbps of dummy bits so all three channels run at 50 kbps

Reading the figure. Channels A and B run at 50 kbps, C at 46 kbps. Four kilobits per second of dummy bits (red) are added to C, so every slot of the frame runs at 50 kbps; now and then C's slot carries a dummy bit (the white hatched slot x). The link carries $3 \times 50 = 150$ kbps.

- Also called **bit padding** or **bit stuffing**.
- **Cost:** link capacity is wasted on the dummy bits.
- It is used in digital hierarchies (for example T-carrier and SONET) to absorb small rate differences between clocks.

(ii) Multiple-slot (multi-slot) multiplexing

When one input's rate is an **integer multiple** of the others, that input is given **more than one slot** per frame. A serial-to-parallel converter splits the fast stream into several streams of the base rate, each of which gets its own slot.

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Figure 21: Multiple-slot allocation: B (10 kbps) gets two slots and C (15 kbps) three slots of 5 kbps per frame (reused from the 2021-22 paper)

Reading the figure. Channel A (5 kbps, blue) gets one slot per frame, B (10 kbps, orange) two and C (15 kbps, green) three; each frame has six 5-kbps slots, so the link runs at $6 \times 5 = 30$ kbps.

- Example (Forouzan): a 50-kbps input and four 25-kbps inputs: the 50-kbps line is split into two 25-kbps streams, giving 6 slots of 25 kbps per frame.
- **Advantage:** no capacity wasted; **limit:** the rates must be integer multiples of a common base rate.

3. Notes and Remarks

- Forouzan, Sec. 6.2 (data rate management). The third method, multilevel multiplexing, combines slow inputs first into a stream equal to a faster input.

4. Final Concise Answer

(a) Serial-to-parallel; odd bits on $\cos \omega_c t$, even bits on $\sin \omega_c t$ (polar, $2T_b$); summed: four phases. Half the bandwidth of BPSK for the same bit rate, the same bit error rate, constant envelope.

(b) Output 11001000 00101101 10100111; 50 000 frames/s; 20 μs; 400 kbps; 2.5 μs.

(c) Each signal on its own carrier, side by side with guard bands, combined on one link; BPFs and demodulators separate them; group, supergroup, master group; radio and TV.

(d) (i) Dummy bits pad slower inputs up to the fastest rate when the rates are not integer multiples; the receiver discards them. (ii) An input whose rate is a multiple of the base rate gets several slots per frame.

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FORMULA SHEET

EEE329

Basic Communication Engineering

Formula Sheet

Session 2019–20 paper

Every formula behind the 3rd Year 2nd Semester Final Examination-2022

TOPICS

1 Information, capacity and noise — 9 formulas

2 Amplitude modulation — 8 formulas

3 Angle modulation — 10 formulas

4 Sampling, quantization and delta modulation — 9 formulas

5 Digital modulation and line codes — 8 formulas

6 Multiplexing — 6 formulas

Each topic is one table: **Title** · **Equation** · **Notation** · **Reference**.

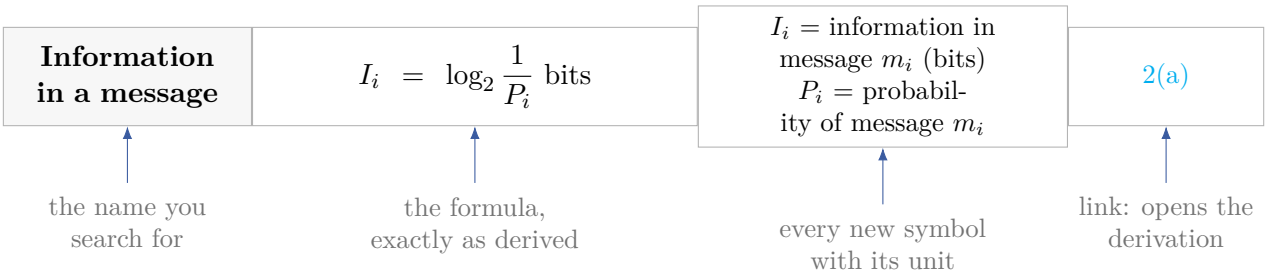
Prepared by Nur Mohammad

November 2026

BACK OF THE COVER

Before you use the sheet

How to read a row



In the exam

- Write the formula first, then put the numbers in: the formula alone often earns a mark.
- Read the symbols in the Notation column before you substitute, and check the units.
- Not sure where a formula comes from? Its Reference opens the derivation.
- The **Key formulas** page at the end gathers the rows marked ★.
- Lost a symbol? The **Symbol index** at the end links to the row where it first appears.

Contents

1 Information, capacity and noise 4

2 Amplitude modulation 5

3 Angle modulation 5

4 Sampling, quantization and delta modulation 6

5 Digital modulation and line codes 7

6 Multiplexing 8

7 Key formulas 9

8 Symbol index 10

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Conventions (Lathi and Ding, 4th ed.): $\omega = 2\pi f$; $m(t)$ is the message, m_p its peak value and B its bandwidth (Hz); β is the FM index; $\log_2$ gives bits; S/N inside a formula is the plain power ratio, not dB. References point to the solutions of the 2019–20 paper.

Columns: Title · Equation · Notation · Ref. (reference).

1. Information, capacity and noise:

Title	Equation	Notation	Ref.
Information in a message	$I_i = \log_2 \dfrac{1}{P_i}$ bits	I_i = information in message m_i (bits) P_i = probability of message m_i	2(a)
★ **Entropy**	$H(m) = \sum_{i=1}^{n} P_i \log_2 \dfrac{1}{P_i}$ bits/message	$H(m)$ = entropy, the average information per message (bits/message) n = number of different messages	2(a)
Count in a long sequence	$N_i = NP_i$	N_i = number of times message m_i occurs N = number of messages sent	2(a)
Information rate	$R = rH$ bits/s	R = information rate (bits/s) r = messages sent per second	2(a)
★ **Shannon capacity**	$C = B \log_2 \left(1 + \dfrac{S}{N}\right)$ bit/s	C = channel capacity, the largest error-free rate (bit/s) B = bandwidth (Hz) S = signal power (W) N = noise power (W) (here, not a count)	1(c)
SNR for a rate	$\dfrac{S}{N} = 2^{C/B} - 1$	doubling C: $1 + \text{SNR}' = (1 + \text{SNR})^2$	1(c)
dB to ratio	$\text{SNR} = 10^{\text{SNR}_{dB}/10}$		1(c)
Thermal noise	$\overline{v_n^2} = 4kTBR$	$\overline{v_n^2}$ = mean-square noise voltage (V^2) k = Boltzmann's constant, 1.38×10^{-23} J/K T = temperature (K) R = resistance (Ω) (here, not a rate)	1(d)
★ **Cascade noise resistance**	$R_{eq} = R_1 + \dfrac{R_2}{A_1^2} + \dfrac{R_3}{A_1^2 A_2^2} + \cdots$	R_{eq} = equivalent noise resistance at the first input (Ω) R_1, R_2, R_3 = noise resistances at the inputs of stages 1, 2, 3 (Ω) A_1, A_2 = voltage gains of stages 1, 2	1(d)

Table 1: Information, capacity and noise

2. Amplitude modulation:

Title	Equation	Notation	Ref.
Product to sum	$\cos A \cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$	A, B = any two angles (rad)	1(b)
★ **DSB-SC of a tone**	$a \cos \omega_m t \cos \omega_c t = \frac{a}{2}\cos(\omega_c - \omega_m)t + \frac{a}{2}\cos(\omega_c + \omega_m)t$	a = message amplitude (V) ω_m = message angular frequency (rad/s) t = time (s) ω_c = carrier angular frequency (rad/s) LSB at $f_c - f_m$, USB at $f_c + f_m$, each $a/2$	1(b)
Square of a sine	$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$	x = any angle (rad)	1(b)
rad/s to Hz	$f = \omega/2\pi$	f = frequency (Hz) ω = angular frequency (rad/s) 377 rad/s = 60 Hz	1(b)
★ **Coherent detector output**	$e_o(t) = \frac{1}{2}m(t)\cos(\Delta\omega t + \delta)$	$e_o(t)$ = detector output (V) $m(t)$ = message signal (V) $\Delta\omega$ = LO frequency error (rad/s) δ = LO phase error (rad)	2(c)
Sideband gap for filtering	$\text{gap} = 2f_L$	f_L = lowest message frequency (Hz)	2(d)
★ **SSB (phase shift)**	$\varphi_{SSB}(t) = m(t)\cos\omega_c t \mp m_h(t)\sin\omega_c t$	$\varphi_{SSB}(t)$ = SSB signal (V) $m_h(t)$ = Hilbert transform of $m(t)$ $-$: USB, $+$: LSB	3(a)
Hilbert transformer	$H(f) = -j\,\text{sgn}(f)$	$H(f)$ = transfer function of the Hilbert transformer j = imaginary unit, $\sqrt{-1}$ $\cos\omega_m t \to \sin\omega_m t$	3(a)

Table 2: Amplitude modulation

3. Angle modulation:

Title	Equation	Notation	Ref.
Instantaneous frequency	$\omega_i = \dfrac{d\theta}{dt}$, $f_i = f_c + \dfrac{1}{2\pi}\dfrac{d\phi}{dt}$	ω_i = instantaneous angular frequency (rad/s) $\theta(t)$ = carrier angle (rad) t = time (s) f_i = instantaneous frequency (Hz) f_c = carrier frequency (Hz) $\phi(t)$ = phase deviation (rad)	2(b)
Power of angle modulation	$P = \dfrac{A^2}{2}$	P = power of the modulated signal (W, 1 Ω load) A = carrier amplitude (V) constant envelope	2(b)

Title	Equation	Notation	Ref.
★ **FM deviation**	$\Delta f = k_f V_m$	Δf = peak frequency deviation (Hz) k_f = frequency sensitivity (Hz/V) V_m = message peak amplitude (V)	4(a)
★ **Modulation index**	$\beta = \dfrac{\Delta f}{f_m}$ (or $\Delta f/B$)	β = FM modulation index f_m = tone frequency (Hz) B = highest message frequency (Hz)	4(a)
★ **Carson's rule**	$B_T = 2(\Delta f + B) = 2B(\beta + 1)$	B_T = transmission bandwidth (Hz)	2(b)
FM line spectrum	$A\sum_n J_n(\beta)\cos(\omega_c + n\omega_m)t$	$J_n(\beta)$ = Bessel function of the first kind, order n n = sideband order ω_c = carrier angular frequency (rad/s) ω_m = tone angular frequency (rad/s) lines f_m apart	3(b)
Multiplier	$f_c \to nf_c$, $\Delta f \to n\Delta f$	n = multiplication factor (here, not a sideband order)	3(c)
Mixer	$f_c \to f_c \pm f_{LO}$, Δf unchanged	f_{LO} = local-oscillator frequency (Hz)	3(c)
★ **Armstrong design**	$n_1 n_2 = \dfrac{\Delta f}{\Delta f_1}$, $f_{LO} = \left\lvert \dfrac{f_c}{n_2} - n_1 f_{c1} \right\rvert$	n_1, n_2 = multiplication factors before and after the mixer Δf_1 = deviation of the NBFM generator (Hz) f_{c1} = carrier of the NBFM generator (Hz) $n_1 = 16$, $n_2 = 128$, $f_{LO} = 440.156$ kHz	3(c)
Slope (differentiator) detector	$\dot{\varphi}_{FM} = -A[\omega_c + k_f m(t)]\sin(\cdot)$	φ_{FM} = FM signal (V) $\dot{\varphi}_{FM}$ = its time derivative $m(t)$ = message signal (V) envelope $\propto m(t)$	4(b)

Table 3: Angle modulation

4. Sampling, quantization and delta modulation:

Title	Equation	Notation	Ref.
Step size	$\Delta = \dfrac{2m_p}{L}$	Δ = quantization step size (V) m_p = message peak value (V) L = number of levels	4(c)
Quantization noise	$\dfrac{\Delta^2}{12} = \dfrac{m_p^2}{3L^2}$	error $\leq \Delta/2$	4(c)
★ **SQNR (uniform)**	$\dfrac{S_o}{N_o} = 3L^2\dfrac{\overline{m^2}}{m_p^2}$	S_o, N_o = output signal and quantization-noise powers (W) $\overline{m^2}$ = mean-square value of the message (V^2)	4(c)
μ-law compressor	$y = \dfrac{\ln(1+\mu\lvert m\rvert/m_p)}{\ln(1+\mu)}\operatorname{sgn}(m)$	y = compressor output (normalized) μ = compression parameter, 255 here	4(c)

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Title	Equation	Notation	Ref.
Sampling rate	$f_s = (1+x)\,2B$	f_s = sampling rate (samples/s) x = fraction above the Nyquist rate B = message bandwidth (Hz)	4(d)
★ **PCM bit rate**	$R = (1+e)\,N_{sig} f_s n$	R = PCM bit rate (bps) e = framing overhead fraction N_{sig} = number of signals multiplexed n = bits per sample	4(d)
Minimum bandwidth	$B_T = R/2$	B_T = minimum transmission bandwidth (Hz)	4(d)
DM update	$\hat{m}_q[k] = \hat{m}_q[k-1] + d_q[k],\ d_q = \pm\sigma$	$\hat{m}_q[k]$ = receiver's staircase estimate at sample k $d_q[k]$ = quantized difference, $\pm\sigma$ σ = DM step size (V)	4(e)
★ **DM receiver**	$\hat{m}_q[k] = \sum_{i=0}^{k} d_q[i]$	i = summation index accumulator + LPF	4(e)

Table 4: Sampling, quantization and delta modulation

5. Digital modulation and line codes:

Title	Equation	Notation	Ref.
AMI	$0 \to 0$ V; 1s alternate $\pm V$	V = pulse amplitude (V)	5(b)
Dicode	$0{\to}1$: $-V$; $1{\to}0$: $+V$; no change: 0	course notes' rule	5(b)
★ **DPSK encoder**	$q_k = I_k \oplus q_{k-1}$	q_k = encoded bit k q_{k-1} = previous encoded bit I_k = data bit k data 1 flips the phase	5(c)
DPSK receiver	LPF output $= \frac{1}{2} a_k a_{k-1} A^2$	a_k = sign (±1) of the received pulse k a_{k-1} = sign of the previous pulse A = carrier amplitude (V) $+ \to 0$, $- \to 1$	5(c)
★ **BFSK correlator**	$\int_0^{T_b} A\cos^2 \omega_1 t\, dt \approx \dfrac{AT_b}{2}$	T_b = bit duration (s) ω_1 = angular frequency sent for a 1 (rad/s) t = time (s) other branch ≈ 0 when orthogonal	5(d)
Coherent BFSK error	$P_b = Q\left(\sqrt{E_b/N_0}\right)$	P_b = bit error probability $Q(x)$ = Gaussian tail probability E_b = energy per bit (J) N_0 = noise power spectral density (W/Hz)	5(d)
★ **QPSK**	$\varphi = a_I A\cos\omega_c t + a_Q A \sin\omega_c t$	φ = QPSK signal (V) a_I, a_Q = in-phase and quadrature bits, ±1 ω_c = carrier angular frequency (rad/s) T_s = symbol duration, $2T_b$ (s)	6(a)
BPSK/QPSK error	$P_b = Q\left(\sqrt{2E_b/N_0}\right)$	same for both	6(a)

Table 5: Digital modulation and line codes

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6. Multiplexing:

Title	Equation	Notation	Ref.
★ **Frame rate**	$\frac{\text{channel rate}}{\text{bits per slot}}$		6(b)
Frame and bit duration	$T_f = 1/\text{frame rate}$, $T_b = 1/R$	T_f = frame duration (s) T_b = bit duration (s) R = link bit rate (bps)	6(b)
Link rate	$R =$ frame rate $\times$ bits per frame		6(b)
★ **FDM bandwidth**	$B_{total} = NB_{ch} + (N-1)B_g$	B_{total} = total FDM bandwidth (Hz) N = number of channels B_{ch} = bandwidth of one channel (Hz) B_g = guard band (Hz)	6(c)
Pulse stuffing	pad slow inputs to the fastest rate	dummy bits discarded	6(d)
Multiple slots	slots per frame $= R_i/R_{base}$	R_i = rate of input i (bps) R_{base} = base (slowest) rate (bps)	6(d)

Table 6: Multiplexing

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7. Key formulas:

The formulas marked ★ in the sheet, on one page. Each title links back to its row.

Title	Equation
Entropy	$H(m) = \sum_{i=1}^{n} P_i \log_2 \frac{1}{P_i}$ bits/message
Shannon capacity	$C = B \log_2 \left(1 + \frac{S}{N}\right)$ bit/s
Cascade noise resistance	$R_{eq} = R_1 + \frac{R_2}{A_1^2} + \frac{R_3}{A_1^2 A_2^2} + \cdots$
DSB-SC of a tone	$a \cos \omega_m t \cos \omega_c t = \frac{a}{2} \cos(\omega_c - \omega_m)t + \frac{a}{2} \cos(\omega_c + \omega_m)t$
Coherent detector output	$e_o(t) = \frac{1}{2} m(t) \cos(\Delta\omega t + \delta)$
SSB (phase shift)	$\varphi_{SSB}(t) = m(t) \cos \omega_c t \mp m_h(t) \sin \omega_c t$
FM deviation	$\Delta f = k_f V_m$
Modulation index	$\beta = \frac{\Delta f}{f_m}$ (or $\Delta f / B$)
Carson's rule	$B_T = 2(\Delta f + B) = 2B(\beta + 1)$
Armstrong design	$n_1 n_2 = \frac{\Delta f}{\Delta f_1}$, $f_{LO} = \left\lvert \frac{f_c}{n_2} - n_1 f_{c1} \right\rvert$
SQNR (uniform)	$\frac{S_o}{N_o} = 3L^2 \frac{\overline{m^2}}{m_p^2}$
PCM bit rate	$R = (1 + e)\, N_{sig} f_s n$
DM receiver	$\hat{m}_q[k] = \sum_{i=0}^{k} d_q[i]$
DPSK encoder	$q_k = I_k \oplus q_{k-1}$
BFSK correlator	$\int_0^{T_b} A \cos^2 \omega_1 t \, dt \approx \frac{A T_b}{2}$
QPSK	$\varphi = a_I A \cos \omega_c t + a_Q A \sin \omega_c t$
Frame rate	$\frac{\text{channel rate}}{\text{bits per slot}}$
FDM bandwidth	$B_{total} = N B_{ch} + (N - 1) B_g$

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8. Symbol index:

Every symbol the sheet defines, in the order it first appears. The last column links to that row.

Symbol	Meaning	First used in
I_i	information in message m_i (bits)	Information in a message
P_i	probability of message m_i	Information in a message
$H(m)$	entropy, the average information per message (bits/message)	Entropy
n	number of different messages	Entropy
N_i	number of times message m_i occurs	Count in a long sequence
N	number of messages sent	Count in a long sequence
R	information rate (bits/s)	Information rate
r	messages sent per second	Information rate
C	channel capacity, the largest error-free rate (bit/s)	Shannon capacity
B	bandwidth (Hz)	Shannon capacity
S	signal power (W)	Shannon capacity
N	noise power (W) (here, not a count)	Shannon capacity
$\overline{v_n^2}$	mean-square noise voltage (V^2)	Thermal noise
k	Boltzmann's constant, 1.38×10^{-23} J/K	Thermal noise
T	temperature (K)	Thermal noise
R	resistance (Ω) (here, not a rate)	Thermal noise
R_{eq}	equivalent noise resistance at the first input (Ω)	Cascade noise resistance
R_1, R_2, R_3	noise resistances at the inputs of stages 1, 2, 3 (Ω)	Cascade noise resistance
A_1, A_2	voltage gains of stages 1, 2	Cascade noise resistance
A, B	any two angles (rad)	Product to sum
a	message amplitude (V)	DSB-SC of a tone
ω_m	message angular frequency (rad/s)	DSB-SC of a tone
t	time (s)	DSB-SC of a tone
ω_c	carrier angular frequency (rad/s)	DSB-SC of a tone
x	any angle (rad)	Square of a sine
f	frequency (Hz)	rad/s to Hz
ω	angular frequency (rad/s)	rad/s to Hz
377 rad/s	60 Hz	rad/s to Hz
$e_o(t)$	detector output (V)	Coherent detector output
$m(t)$	message signal (V)	Coherent detector output
$\Delta\omega$	LO frequency error (rad/s)	Coherent detector output
δ	LO phase error (rad)	Coherent detector output
f_L	lowest message frequency (Hz)	Sideband gap for filtering
$\varphi_{SSB}(t)$	SSB signal (V)	SSB (phase shift)
$m_h(t)$	Hilbert transform of $m(t)$	SSB (phase shift)
$H(f)$	transfer function of the Hilbert transformer	Hilbert transformer

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Symbol	Meaning	First used in
j	imaginary unit, $\sqrt{-1}$	Hilbert transformer
ω_i	instantaneous angular frequency (rad/s)	Instantaneous frequency
$\theta(t)$	carrier angle (rad)	Instantaneous frequency
f_i	instantaneous frequency (Hz)	Instantaneous frequency
f_c	carrier frequency (Hz)	Instantaneous frequency
$\phi(t)$	phase deviation (rad)	Instantaneous frequency
P	power of the modulated signal (W, 1 Ω load)	Power of angle modulation
A	carrier amplitude (V)	Power of angle modulation
Δf	peak frequency deviation (Hz)	FM deviation
k_f	frequency sensitivity (Hz/V)	FM deviation
V_m	message peak amplitude (V)	FM deviation
β	FM modulation index	Modulation index
f_m	tone frequency (Hz)	Modulation index
B	highest message frequency (Hz)	Modulation index
B_T	transmission bandwidth (Hz)	Carson's rule
$J_n(\beta)$	Bessel function of the first kind, order n	FM line spectrum
n	sideband order	FM line spectrum
ω_m	tone angular frequency (rad/s)	FM line spectrum
n	multiplication factor (here, not a sideband order)	Multiplier
f_{LO}	local-oscillator frequency (Hz)	Mixer
n_1, n_2	multiplication factors before and after the mixer	Armstrong design
Δf_1	deviation of the NBFM generator (Hz)	Armstrong design
f_{c1}	carrier of the NBFM generator (Hz)	Armstrong design
φ_{FM}	FM signal (V)	Slope (differentiator) detector
$\dot{\varphi}_{FM}$	its time derivative	Slope (differentiator) detector
Δ	quantization step size (V)	Step size
m_p	message peak value (V)	Step size
L	number of levels	Step size
S_o, N_o	output signal and quantization-noise powers (W)	SQNR (uniform)
$\overline{m^2}$	mean-square value of the message (V^2)	SQNR (uniform)
y	compressor output (normalized)	μ-law compressor
μ	compression parameter, 255 here	μ-law compressor
f_s	sampling rate (samples/s)	Sampling rate
x	fraction above the Nyquist rate	Sampling rate
R	PCM bit rate (bps)	PCM bit rate
e	framing overhead fraction	PCM bit rate
N_{sig}	number of signals multiplexed	PCM bit rate

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Symbol	Meaning	First used in
n	bits per sample	PCM bit rate
$\hat{m}_q[k]$	receiver's staircase estimate at sample k	DM update
$d_q[k]$	quantized difference, $\pm\sigma$	DM update
σ	DM step size (V)	DM update
i	summation index	DM receiver
V	pulse amplitude (V)	AMI
q_k	encoded bit k	DPSK encoder
q_{k-1}	previous encoded bit	DPSK encoder
I_k	data bit k	DPSK encoder
a_k	sign (±1) of the received pulse k	DPSK receiver
a_{k-1}	sign of the previous pulse	DPSK receiver
T_b	bit duration (s)	BFSK correlator
ω_1	angular frequency sent for a 1 (rad/s)	BFSK correlator
P_b	bit error probability	Coherent BFSK error
$Q(x)$	Gaussian tail probability	Coherent BFSK error
E_b	energy per bit (J)	Coherent BFSK error
N_0	noise power spectral density (W/Hz)	Coherent BFSK error
φ	QPSK signal (V)	QPSK
a_I, a_Q	in-phase and quadrature bits, ±1	QPSK
T_s	symbol duration, $2T_b$ (s)	QPSK
T_f	frame duration (s)	Frame and bit duration
R	link bit rate (bps)	Frame and bit duration
B_{total}	total FDM bandwidth (Hz)	FDM bandwidth
N	number of channels	FDM bandwidth
B_{ch}	bandwidth of one channel (Hz)	FDM bandwidth
B_g	guard band (Hz)	FDM bandwidth
R_i	rate of input i (bps)	Multiple slots
R_{base}	base (slowest) rate (bps)	Multiple slots

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